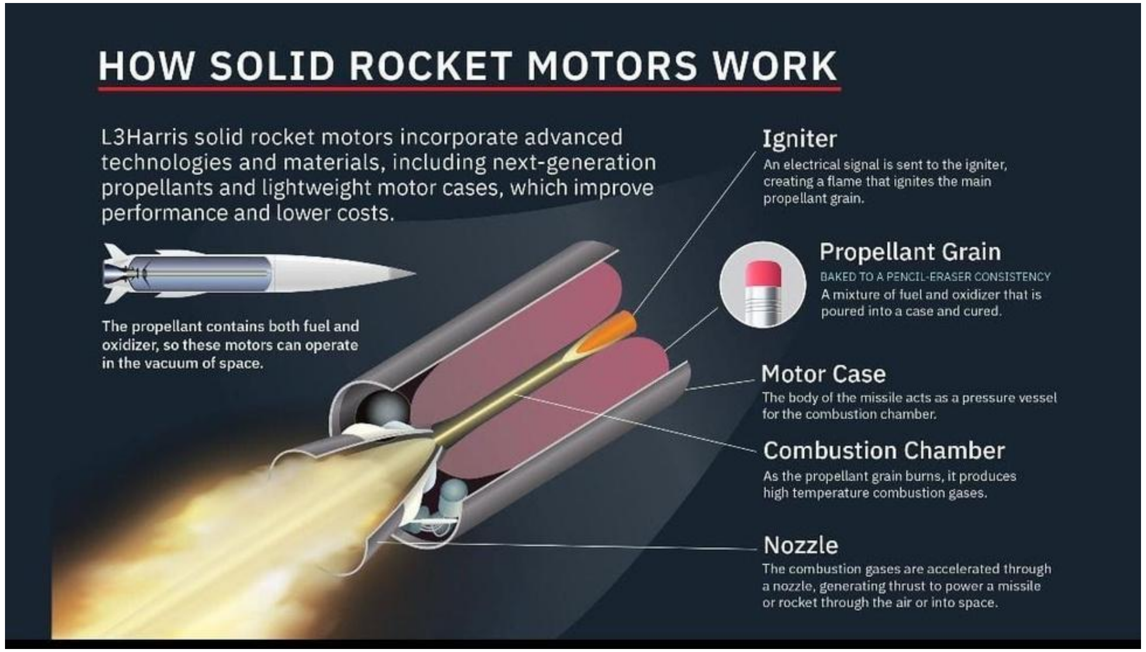
Motor Selection

Propulsion Subsystem Design — Motor Selection: ASTROFORGE VAYU is a PVC/Fiberglass composite airframe, placing it in the PVC body class (11.5 kg liftoff mass cap, ~3 kg motor mass) per competition Requirement 1.2.

Parameter	Motor A — TIVRA K-925	Motor B — TIVRA K-560
Total Impulse	2500 Ns	1960 Ns
Average Thrust	704 N	441 N
Peak Thrust	1559 N	1030 N
Burn Time	3.6 s	4.35 s
Motor Weight	4459 g	3710 g
Vehicle Liftoff Weight	15,000 g	11,500 g

Motor Selection

- Motor A (K-925) matches the Aluminum body class (15,000 g), not this PVC/Fiberglass vehicle — using it would exceed the PVC class cap by ~3.5 kg and violate Requirement 1.2.
- Motor B (K-560) matches the PVC body class liftoff cap of 11,500 g exactly.
- **Selected Motor: TIVRA K-560 (Part No. K-560, Impulse Class K, 1960 Ns — within the 2800 Ns competition cap per Req. 1.7)**



Mass Budget Reconciliation

Propulsion Subsystem Design — Mass Budget Reconciliation: The PDR-stage mass budget totaled 6.6 kg liftoff mass. With the TIVRA K-560 selected, actual vehicle liftoff mass is 11.5 kg — the full PVC-class allowance. This must be propagated through every mass-dependent analysis before CDR submission:

- Stability margin (CG/CP) — must be recalculated at 11.5 kg loaded mass.
- Altitude / apogee simulation (OpenRocket or equivalent) — must be rerun at 11.5 kg.
- Descent rate and recovery sizing — descent mass changes once motor casing mass is subtracted post-burnout.
- Structural load paths — airframe re-checked for higher inertial loads at 11.5 kg under 15g launch acceleration and 30g shock (Req. 3.1).

This reconciliation is flagged as an action item for the Rocket Architecture and Recovery sub-teams ahead of CDR submission.

Thrust-to-Weight Analysis

Propulsion Subsystem Design — Thrust-to-Weight Analysis

Parameter	Formula	Value
Vehicle Weight (W)	$W = m \cdot g = 11.5\ kg \times 9.81\ m/s^2$	112.8 N
T/W — Average Thrust	441 N / 112.8 N	3.9
T/W — Peak Thrust	1030 N / 112.8 N	9.1

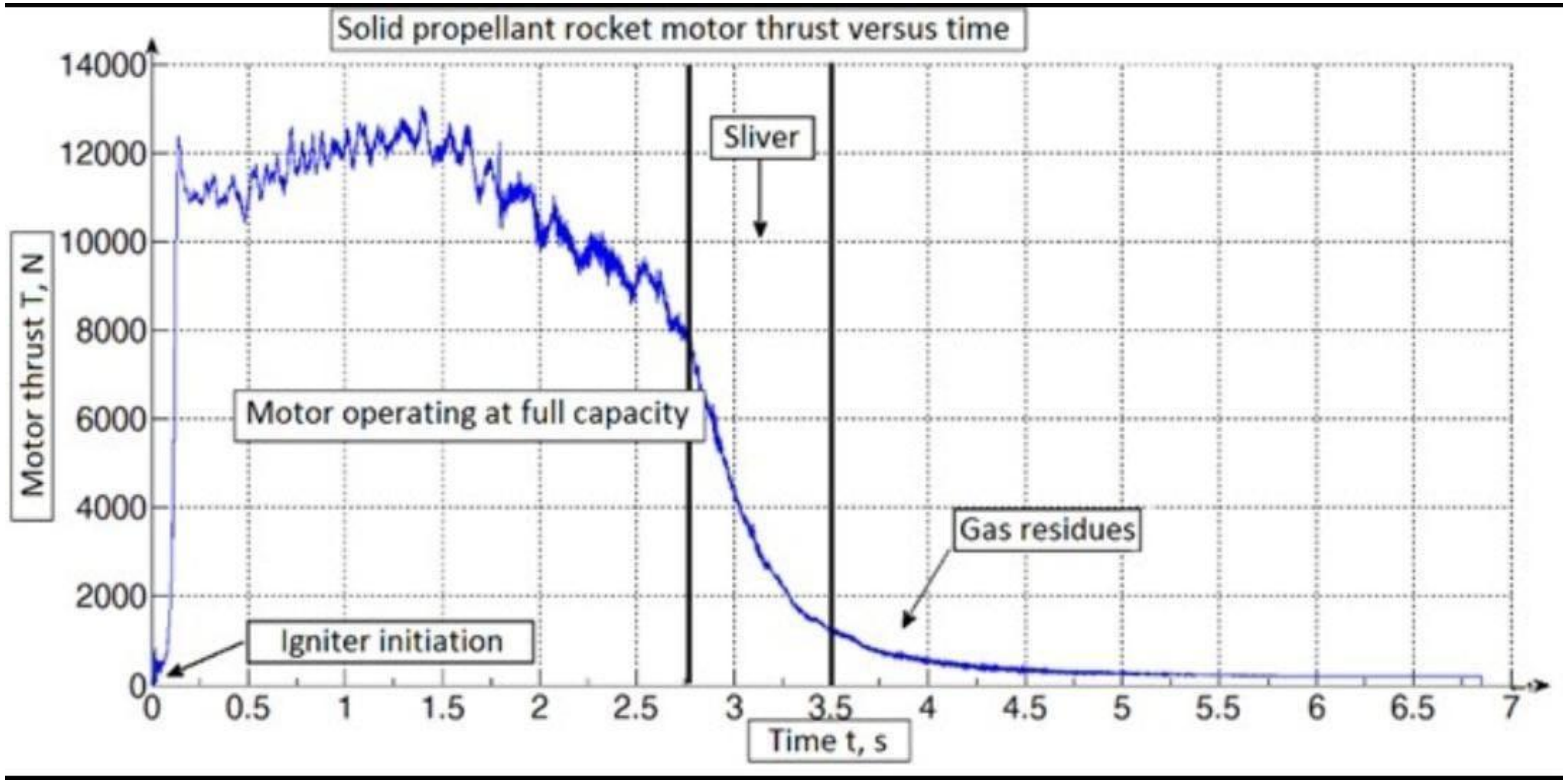
Thrust-to-Weight Analysis

- A minimum average thrust-to-weight ratio of approximately 5:1 is generally recommended for a clean, stable rail departure in model rocketry.

- The K-560's average T/W of 3.9 falls below this guideline. This does not disqualify the motor, but rail exit velocity must be verified explicitly against the team's launch rail length to confirm adequate velocity for aerodynamic stabilization (fin authority) at rail departure.

- **This check is carried forward as an open action item for the CDR stability section.**

Thrust-to-Weight Analysis



Thrust Generation Mechanism

Propulsion Subsystem Design — Thrust Generation Mechanism

- Combustion & Gas Generation: ignition initiates burning along the exposed grain surface, converting stored chemical energy into high-pressure, high-temperature combustion gas.
- Nozzle Expansion: gas is exhausted through a de Laval (converging-diverging) nozzle, accelerating flow to supersonic velocity and converting thermal/pressure energy into kinetic energy.
- Thrust Generation (Newton's 3rd Law): momentum imparted to the exhaust jet produces an equal and opposite reaction force on the motor casing, transmitted through the phenolic motor mount into the main airframe.
- Thrust Profile: peak thrust (1030 N) occurs early when burn surface area is largest; thrust decays toward the 441 N average over the 4.35 s burn — a regressive burn profile.
- Powered Ascent to Burnout: thrust must exceed weight (112.8 N) plus drag for positive acceleration; at burnout the vehicle coasts ballistically to apogee.

Data Handoff to Sub-Teams

Propulsion Subsystem Design — Data Handoff to Sub-Teams:

- To Rocket Architecture & Design (enclosure / motor mount sizing)

Parameter	Value
Motor	TIVRA K-560
Motor Loaded Mass	3710 g
Total Vehicle Liftoff Mass	11,500 g (11.5 kg)
Peak Thrust (structural load case)	1030 N

Note: exact propellant mass fraction should be confirmed against the TIVRA K-560 datasheet so Recovery can compute precise burnout mass for descent rate sizing (Req. 7.2: 2–5 m/s ± 0.5 m/s).

Data Handoff to Sub-Teams

Propulsion Subsystem Design — Data Handoff to Sub-Teams:

- To Descent & Recovery Architecture (descent mass /

Parameter	Value
Burnout Time	4.35 s after ignition
Total Impulse	1960 Ns
Approx. Post-Burn Mass	11,500 g minus propellant mass consumed (3710 g motor casing remains onboard through descent)

Forward Interface Requirements

Propulsion Subsystem Design — Forward Interface Requirements

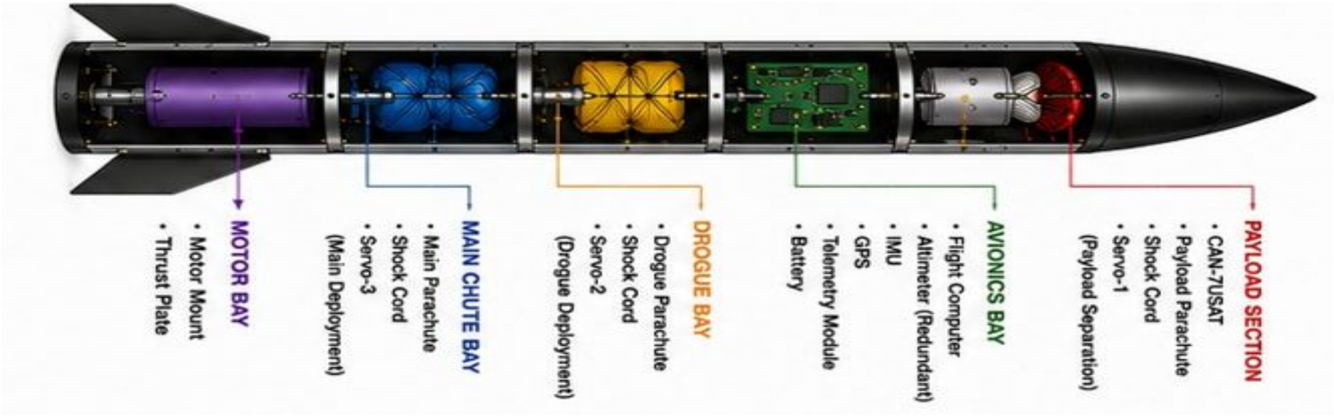
- Motor retention interface requirements issued to Rocket Architecture & Design, per Requirement 1.8 (positive mechanical retention, removable post-flight).

Interface Parameter	Requirement
Thread Specification	M12 × 1.75, 6H tolerance class
Minimum Thread Engagement	20 mm
Aeropack-to-Motor Clearance	10–20 mm

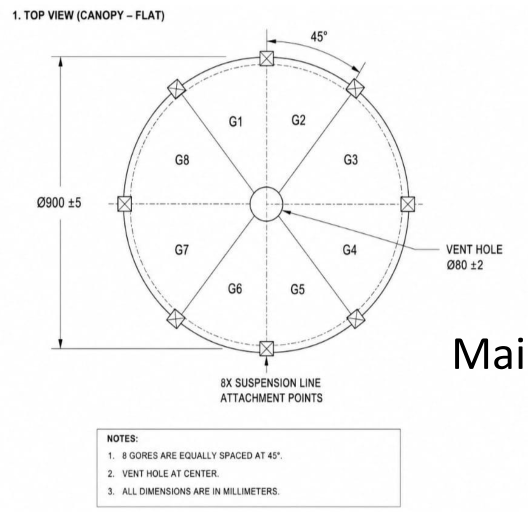
- These values define the retention cap / motor mount tube interface and should be reflected directly in the Architecture sub-team’s motor mount CAD and Bill of Materials for CDR.

CANSAT configuration overview

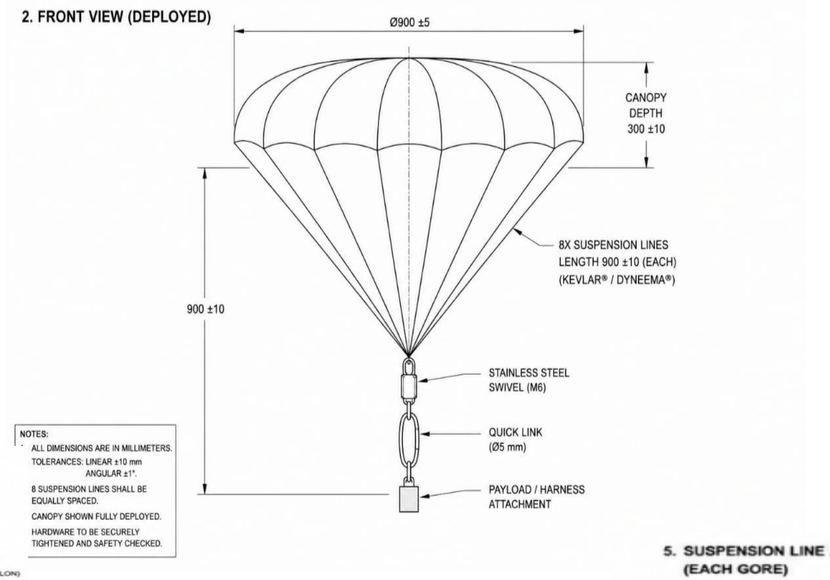
Recovery system

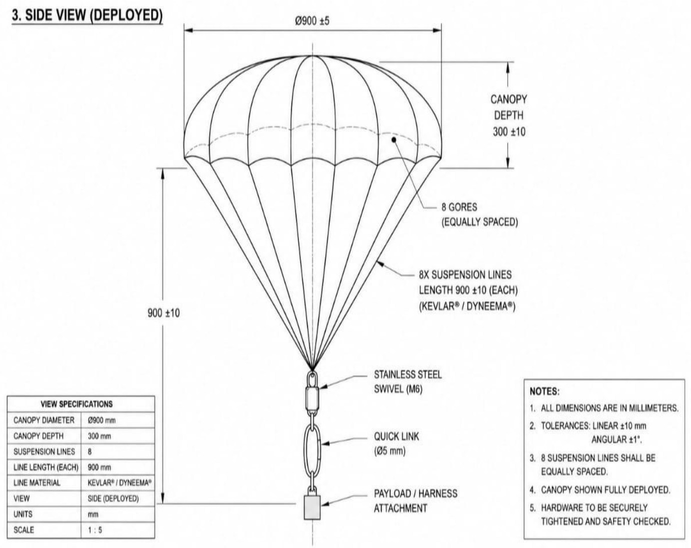


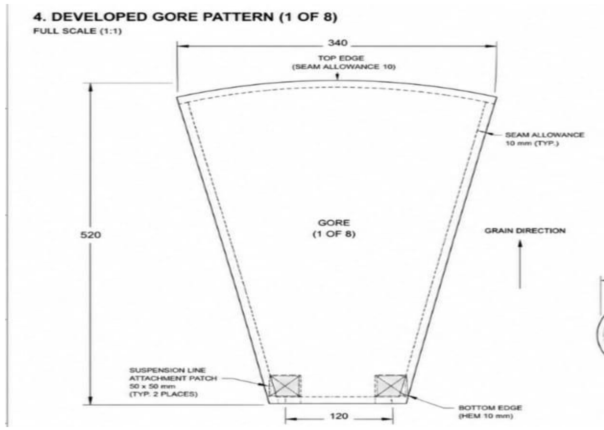
Design of subsystems including ground systems

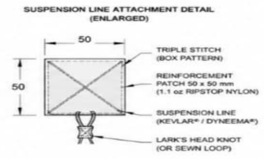


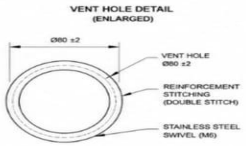
Main parachute



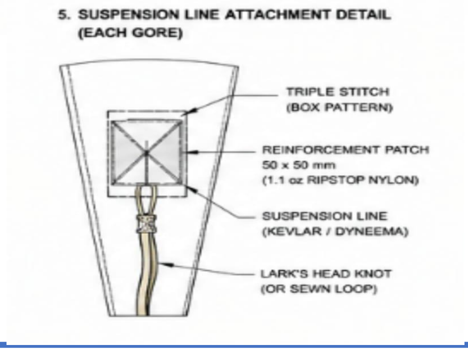






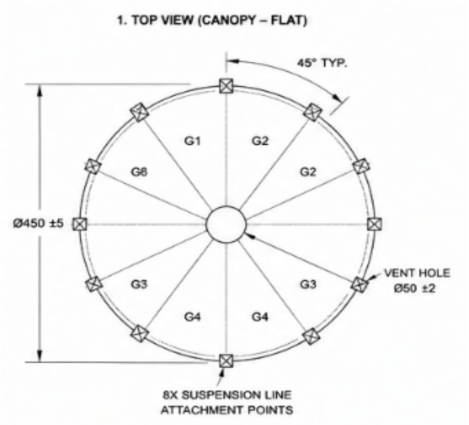


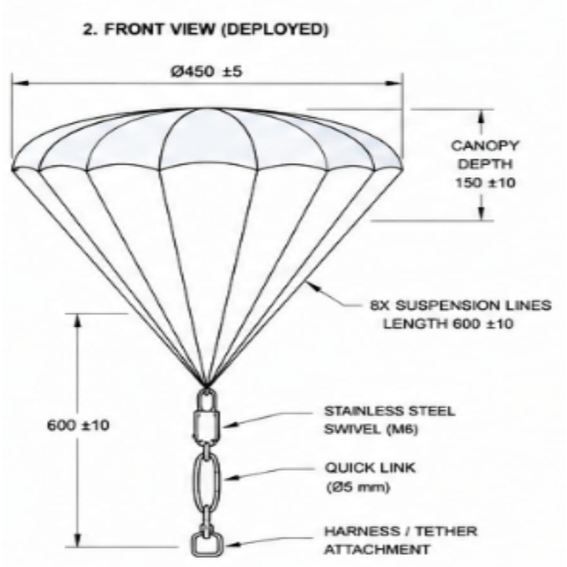
All the measurements in these drawings are in **millimetres (mm)**

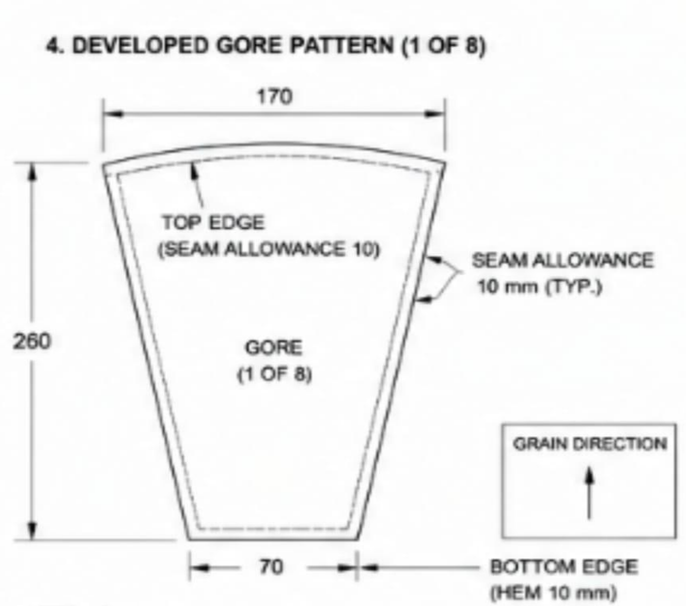


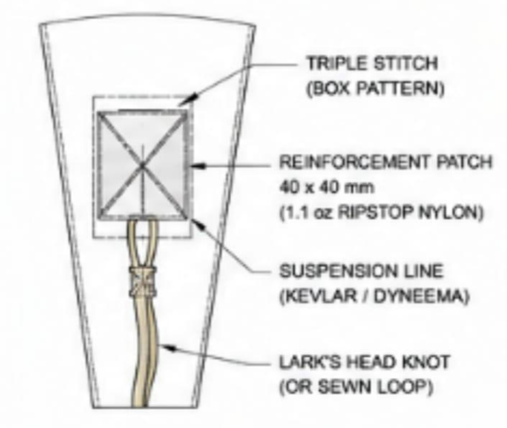
Design of subsystems including ground systems

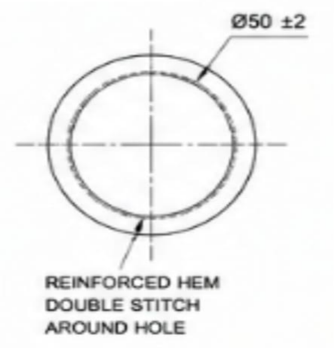
Droguechute
parachute







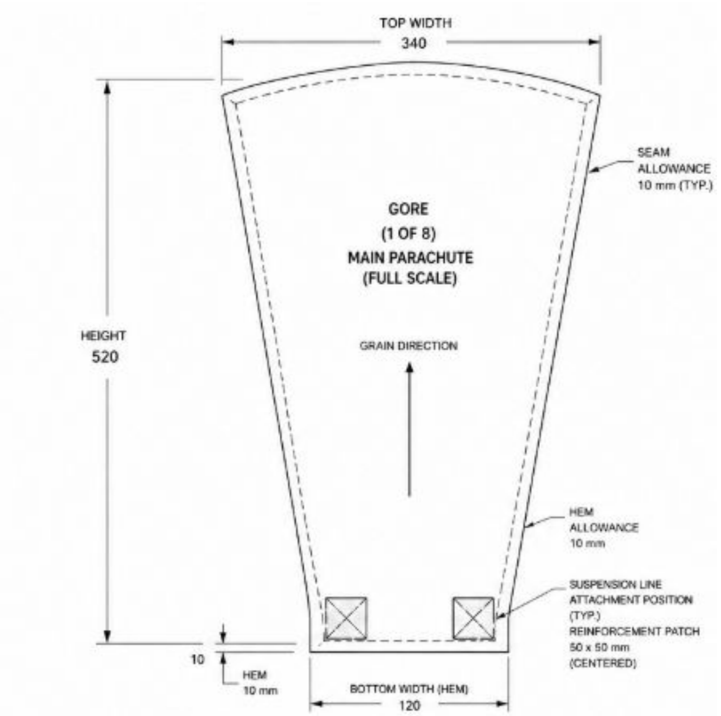


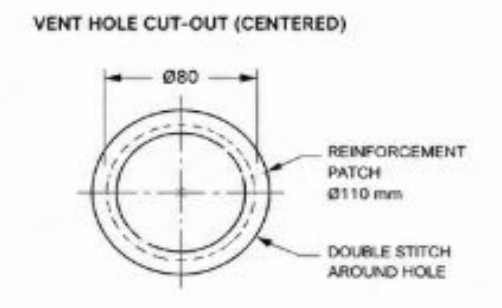


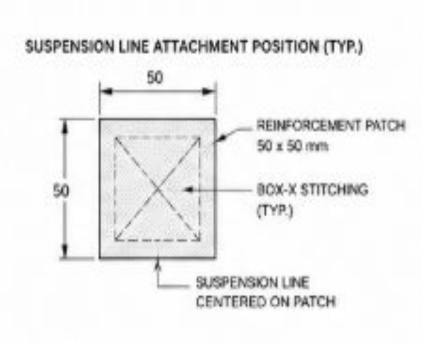
All the measurements in these drawings are in **millimetres (mm)**

Components and hardware realization plan

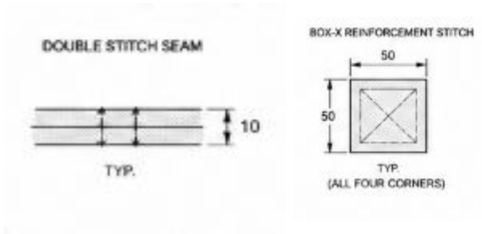
Gore Cutting





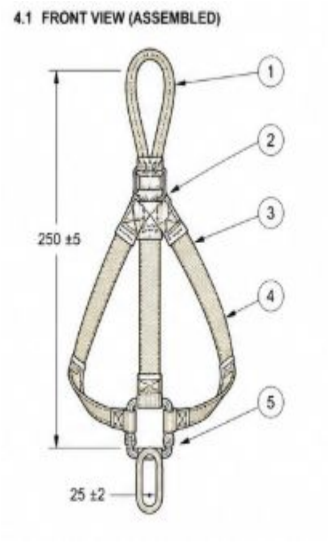


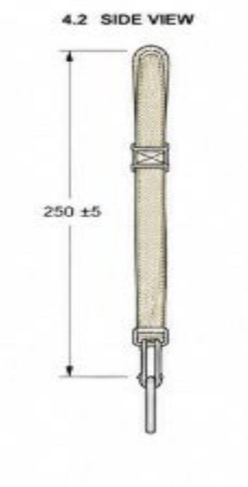
Stitching details

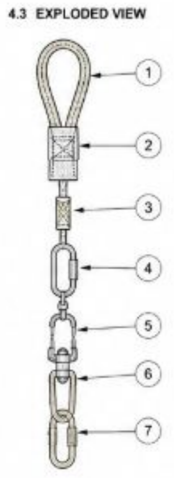


All the measurements in these drawings are in **millimetres (mm)**

Tests undertaken and results

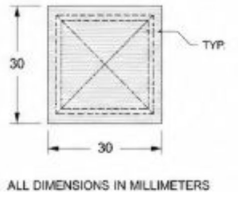




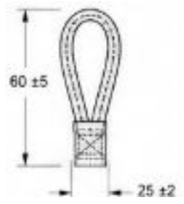


ITEM	DESCRIPTION	MATERIAL	QTY	REMARKS
1	SOFT LINK	NYLON WEBBING	1	25 mm
2	HARNESS WEBBING	NYLON WEBBING	1	25 mm
3	REINFORCED STITCHING	POLYESTER THREAD	1	BOX-X PATTERN
4	QUICK LINK	STAINLESS STEEL	1	Ø5 mm
5	SWIVEL	STAINLESS STEEL	1	MBS ≥ 20 kg
6	SHOCK CORD	ELASTIC LATEX	1	Ø3 mm
7	PAYLOAD ATTACHMENT	STAINLESS STEEL	1	MBS ≥ 20 kg

Stitching details



Soft link details

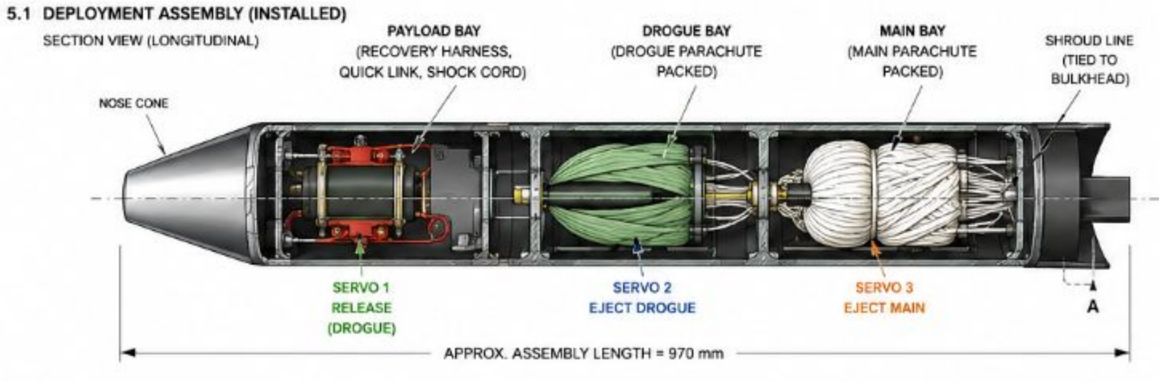


Recovery harness assembly drawing

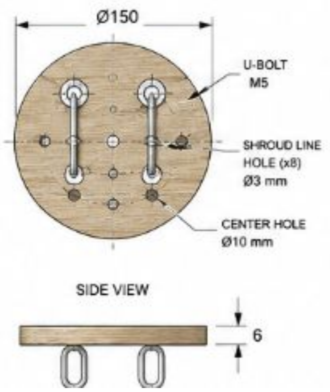
All the measurements in these drawings are in **millimetres (mm)**

CANSAT/Model Rocket mission control software and ground systems

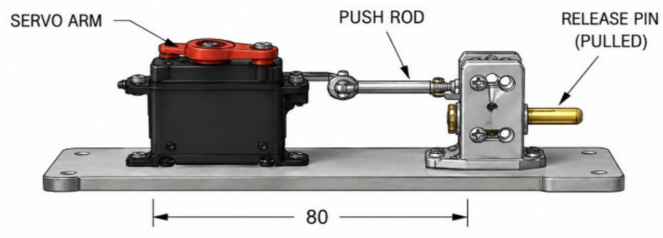
Parachute deployement

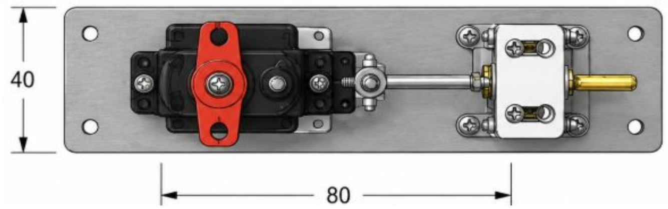


Bulk head



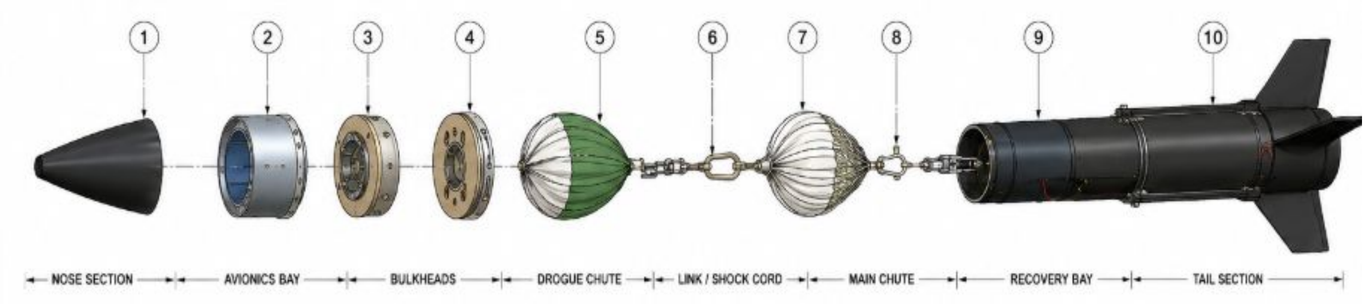
Deployment mechanism





All the measurements in these drawings are in **millimetres (mm)**

Realization and readiness plan including Quality control



SPECIFICATIONS	
TOTAL LENGTH (ASSEMBLED)	970 mm (APPROX.)
RECOVERY SYSTEM MASS	1.25 kg (APPROX.)
DROGUE CHUTE DIAMETER	Ø450 mm
MAIN CHUTE DIAMETER	Ø900 mm
SHOCK CORD LENGTH	100 mm (FREE)
QUICK LINK MBS	≥ 20 kg
RECOVERY SYSTEM	DUAL DEPLOYMENT
COMPLIANCE	NAR / TRA 2024

BILL OF MATERIALS

ITEM	DESCRIPTION	MATERIAL	QTY	REMARKS
1	NOSE CONE	FIBERGLASS / ABS	1	WITH SCREW
2	AVIONICS BAY	ALUMINUM 6061	1	ANODIZED
3	PAYLOAD MOUNT	PLYWOOD (6 mm)	1	WITH U-BOLTS
4	MAIN BULKHEAD	PLYWOOD (6 mm)	1	WITH U-BOLTS
5	DROGUE PARACHUTE (Ø450 mm)	1.1 oz RIPSTOP NYLON	1	WITH LINES
6	RECOVERY HARNESS	KEVLAR® WEBBING	1	ASSEMBLY
7	MAIN PARACHUTE (Ø900 mm)	1.1 oz RIPSTOP NYLON	1	WITH LINES
8	QUICK LINK (MBS ≥ 20 kg)	STAINLESS STEEL	2	CONNECTORS
9	SHOCK CORD (Ø3 mm)	LATEX	1	100 mm (FREE)
10	ROCKET BODY (WITH FINS)	FIBERGLASS / CARBON	1	STRUCTURAL

All the measurements in these drawings are in **millimetres (mm)**

Components and hardware realization plan (Sensors)

1.Barometric Altimeter (Primary)

Model	Image	Interface	Resolution	Weight	Cost
BMP388		[illegible]I	0.08 m	1.5 g	₹400

Reason for Selection:
Best resolution, low power, covers pressure and temp

Image source: Bosch Sensortec / Adafruit / SparkFun.

2. Backup COTS Altimeter

Model	Image	Interface	Resolution	Weight	Cost
Jolly Logic AltimeterTwo		Standalone — independent of flight computer	0.3 m altitude resolution	0.8g	₹3000-₹5000

Reason for Selection: AltimeterTwo chosen over StratoLogger CF for lower mass (8g vs 11g), lower cost, built-in coin cell battery eliminating need for separate 9V battery, built-in LCD display for pad-side verification, and built-in buzzer doubling as secondary recovery aid. Altitude data logging already covered by primary BMP388 and NEO-M10S sensors. Range of 0–9144m is sufficient for 1000m target altitude.

Components and hardware realization plan (Sensors)

3. Tri-axis Accelerometer Low-G

Model	Image	Interface	Resolution	Weight	Cost
ICM-20948		I2C / SPI	Accel: ±2/4/8/16g, Gyro: ±250–4000° /s, Magnetomater range: ±4900 µT (AK09916)	1.5g	₹800-1200

4. High-G Accelerometer

Model	Image	Interface	Resolution	Weight	Cost
H3LIS331DL		I2C / SPI	±100 / ±200 / ±400g selectable range	1.5 g	₹500 – ₹800

Reason for Selection:

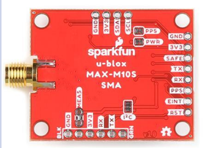
Lower power consumption (0.9mA), SPI support for faster data transfer, onboard FIFO buffer prevents data loss during peak events, better vibration tolerance. Covers orientation and acceleration telemetry fields.

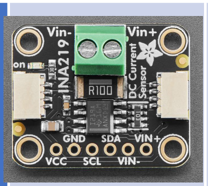
Reason for Selection

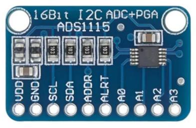
LSM6DS3 maximum range is ±16g which is insufficient margin for 30g shock requirement at separation. H3LIS331DL provides ±400g range giving 13× safety margin over the 30g shock requirement. Purpose-built for high-g shock detection. Digital output eliminates ADC noise issues.

Components and hardware realization plan (Sensors)

5. GNSS with NavIC & 6. Voltage monitor

Model	Image	Interface	Resolution	Weight	Cost
NEO-M10S		UART	Position: 1.5m CEP, Coordinate: 0.0001° (meets telemetry requirement)	12.0 g	₹3,000 – ₹4,000

Model	Image	Interface	Resolution	Weight	Cost
INA219		I2C	Voltage: 4mV, Current: 0.8mA, Power: 0.2mW	1.5 g	₹150 – ₹250

Model	Image	Interface	Resolution	Weight	Cost
ADS1115		I2C	16-bit (0.0001V resolution at ±0.256V range)	1.5 g	₹200 – ₹300

Reason for Selection

Only widely available module with full NavIC support — encouraged by competition guidelines Section 5.1(xvi). Supports GPS + GLONASS + Galileo + BeiDou + NavIC simultaneously. 25Hz update rate, lowest power in class at 18mA, 1.5m position accuracy.

Reason for Selection

Specifically engineered for vibration-heavy platforms including drones and rockets. Superior vibration rejection compared to MPU6050 and LSM6DS3. Used in professional drone flight controllers for this exact reason. Covers gyro spin rate telemetry field. Complements H3LIS331DL — BMI088 covers normal flight dynamics, H3LIS331DL covers shock events

MODEL ROCKETRY INDIA

Components and hardware realization plan (Sensors)

7.Temperature Sensor

Model	Image	Interface	Resolution	Weight	Cost
DS18B20		1-Wire (single data pin)	0.0625°C	3.0 g	₹100 – ₹200

8. Deployment Confirmation Switch

Model	Image	Interface	Resolution	Weight	Cost
Omron D2F-01F		GPIO (digital input)	Binary — 0 (not deployed) / 1 (deployed)	1.6 g	₹80 – ₹150

Reason for Selection: Measures true ambient air temperature independently of PCB heat more accurate than using BMP388 internal temperature for atmospheric readings. 1-Wire interface uses only one GPIO pin. Resolution of 0.0625°C exceeds the 0.1°C telemetry requirement. Wide range of −55°C to +125°C covers all flight conditions. Requires only one 4.7kΩ pull-up resistor.

Reason for Selection

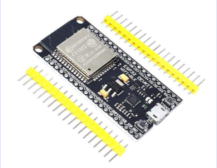
Physically confirms CAN-7USAT ejection has occurred — critical for flight software state transition from PAYLOAD_SEP to DESCENT. Omron D2F series is industry standard subminiature micro switch with proven reliability. Actuation force 0.74N suitable for deployment confirmation. At 1.6g it is the lightest available confirmation method. Simple GPIO connection requires no additional circuitry.

Components and hardware realization plan (Sensors)

9. Recovery Buzzer/Beacon

Model	Image	Interface	Resolution	Weight	Cost
Passive piezo buzzer		GPIO (PWM output)	Not applicable — audible output device	5.0 g	₹20 – ₹50

10. Flight Controler

Model	Image	Interface	Processor	Weight	Cost
ESP32 DevKit V1		Wi-Fi, Bluetooth, UART, SPI, I2C, PWM, ADC	Dual-core Xtensa LX6 - 240 MHz	10.0 g	₹350 – ₹700

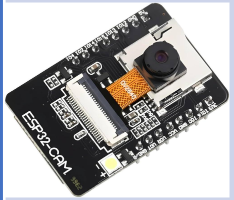
Reason for Selection

Activates on IMPACT state to aid physical recovery of rocket in open field. Passive piezo chosen over active buzzer for frequency control via PWM — allows distinctive beep patterns. 85–95 dB output audible at 50m+ in open field. Near-zero cost and mass. Aligned with optional recovery technique objective in Section 4.2(vii) of competition guidelines.

Reason For Selection: Chosen over STM32 and Arduino Nano because it provides sufficient processing power for simultaneous sensor fusion, telemetry packaging, deployment logic, and data logging in a single low-cost board. Dual-core architecture allows separation of telemetry and flight-critical tasks. Built-in Wi-Fi and Bluetooth simplify debugging and ground testing without extra hardware. Much cheaper than Teensy while still meeting all avionics requirements. Large Arduino/Python ecosystem reduces software development risk and ensures long-term maintainability.

Components and hardware realization plan (Sensors)

11. Camera Module(Optional)

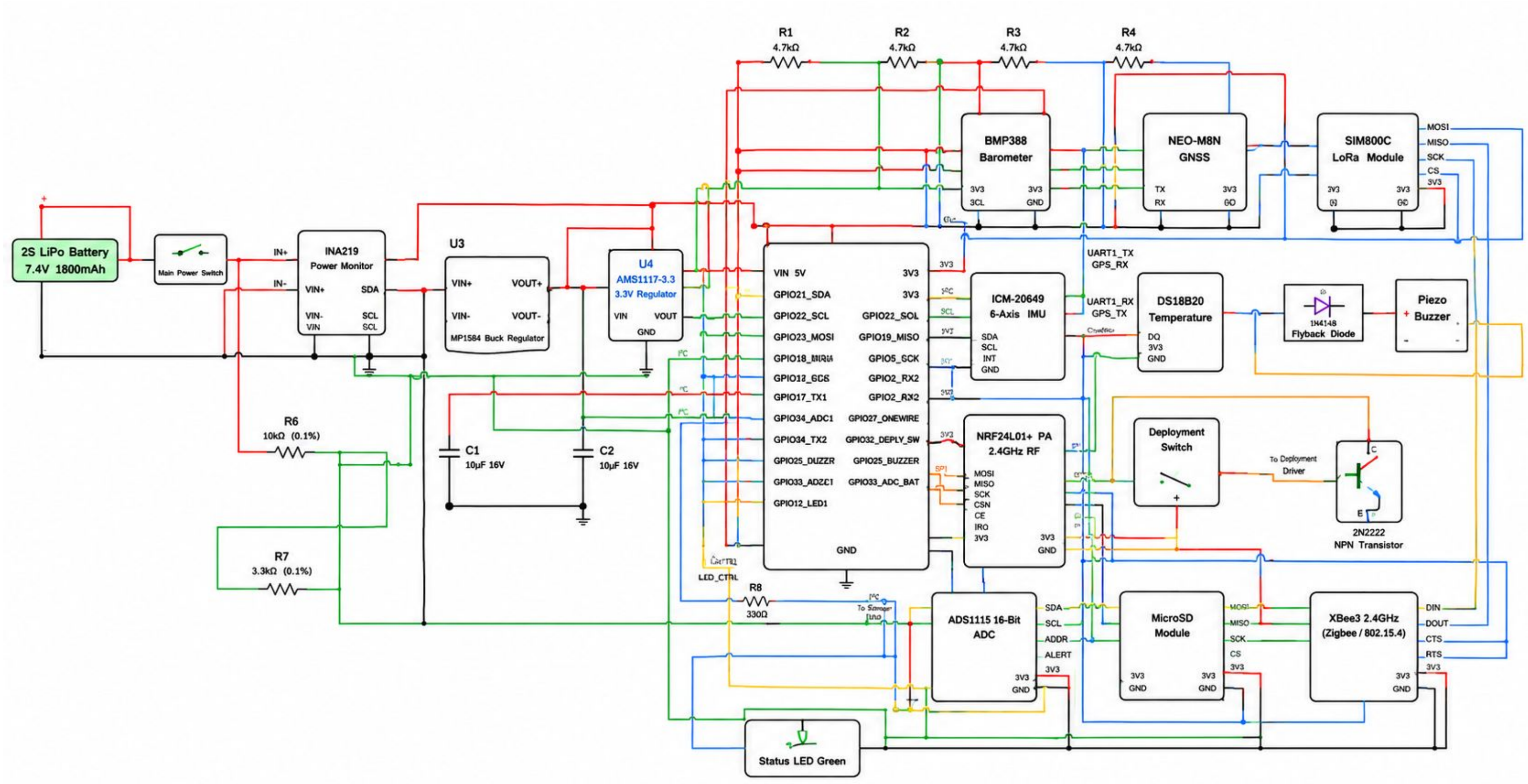
Model	Image	Interface	Resolution	Weight	Cost
ESP32-CAM (OV2640)		SPI Camera Interface / UART Programming	Up to 1600 × 1200 (UXGA)	5.0–8.0 g	₹500 – ₹900

Reason For Selection:

Chosen over Raspberry Pi Camera and GoPro-style cameras because it provides the best mass-to-cost ratio for educational flight imaging. Integrated ESP32 + OV2640 architecture eliminates need for separate SBC or image processor. Built-in microSD support enables onboard image logging without extra hardware. Much lighter and cheaper than action cameras while still capable of capturing launch and recovery imagery. Selected as optional payload only — primary avionics remain fully independent to avoid mission risk

Sensor	Model	Current (mA)	Mass (g)
Flight Computer	ESP32 DevKit V1	80.00	10.0
Barometric Altimeter	BMP388	3.40	1.5
9-DOF IMU	ICM-20948	2.50	1.5
High-G Accelerometer	H3LIS331DL	1.00	1.5
GNSS Module	NEO-M10S	18.00	12.0
Temperature Sensor	DS18B20	1.50	3.0
Voltage Monitor	INA219	1.00	1.5Module (Optional)
ADC	ADS1115	0.15	1.5
Deployment Switch	Omron D2F-01F	0.00	1.6
Recovery Buzzer	Passive Piezo	0.00 (idle)	5.0
SD Card Module	MicroSD SPI	0.00 (idle)	3.0
Voltage Regulator	MP1584 Buck	5.00	3.0
Pull-up Resistors	Passive	0.50	0.5
Camera Module(Optional)	ESP32-CAM (OV2640)	120-180	8.0g
Subtotal (sensors + MCU)		**113.05 mA (+120 mA)**	**46.1 g + 8.0g**

Model Rocket mission control software and ground systems



Model Rocket mission control software and ground systems

Ground Station Overview

Purpose, objectives, functional scope

Communication Architecture

Telemetry link, command link, radio config

Hardware & Software Design

Ground Station components and modules

Sensor & Packet Integration

Onboard sensors, wiring, telemetry structure

Dashboard & Software Workflow

Operator interface and processing pipeline

Testing, Risk & Roadmap

Validation plan, risk mitigation, next steps

Model Rocket mission control software and ground systems

Ground Station Overview

The Ground Station is the primary interface between the rocket and mission operators — receiving telemetry, displaying flight parameters in real time, logging mission data, and monitoring vehicle health throughout flight.

WHAT MAKES IT DIFFERENT

Unlike the onboard flight computer — focused on acquisition and control — the Ground Station focuses on visualization, communication, and post-flight analysis.

OBJECTIVES

- Receive telemetry packets transmitted from the rocket
- Display mission-critical parameters in real time
- Log all received telemetry for post-flight analysis
- Monitor link health and communication quality

CORE RESPONSIBILITIES

Telemetry Reception

Receive packets over the designated radio link

Data Processing

Decode packets into altitude, GPS, voltage, status

Data Visualization

Continuously updating operator dashboard

Data Logging

Structured CSV storage for post-flight analysis

Comm. Monitoring

Link status, packet count, RSSI, error rate

Model Rocket mission control software and ground systems

Rocket Telemetry → RF Link & Ground Radio → Python Ground Station → Dashboard & CSV Logging

DESIGN GOALS

1

Reliability

Continuous telemetry reception during flight

2

Scalability

Easy addition of future sensors and commands

3

Maintainability

Modular software architecture

4

Portability

Runs on any standard Windows laptop

5

Usability

Clear and intuitive operator interface

Model Rocket mission control software and ground systems

ROCKET SIDE

Sensors
(GPS · IMU · Baro · Battery)

↓

Flight Computer
(ESP32)

↓

Radio Module
(XBee / 868 MHz)

RF LINK →

GROUND STATION SIDE

Ground Station Radio

↓

USB Serial Interface

↓

Python Ground Station

↓

Dashboard | CSV Logger | Packet Parser | Link Monitor

COMMUNICATION INTERFACES

UART ESP32 ↔ Radio Module

RF Wireless communication

USB Ground Radio ↔ Laptop

COM Port Python ↔ Radio

Model Rocket mission control software and ground systems

Telemetry and command traffic are physically separated across two radios — improving reliability, reducing interference, and allowing each link to be configured independently.

Telemetry Link

868 MHz

- Sensor data transmission
- Flight status updates
- GPS coordinates
- Battery monitoring
- Flight event reporting

Command Link

2.4 GHz XBee · IEEE 802.15.4

- Ground Station commands
- Configuration updates
- Emergency commands
- Future command & control expansion

Model Rocket mission control software and ground systems

Radio Configuration

The competition specifies radio technology and communication parameters. Before launch, rocket and Ground Station radios are configured with identical settings to establish a compatible link.

XBEE CONFIGURATION — COMPETITION REQUIREMENT

Parameter	Configuration
Radio	XBee 2.4 GHz
Mode	IEEE 802.15.4
Main Channel	Competition Assigned
Backup Channel	Competition Assigned
PAN ID	0x1000 + Team ID

868 MHz TELEMETRY CONFIGURATION

Parameter	Configuration
Main Frequency	Competition Assigned
Backup Frequency	Competition Assigned
Spreading Factor	Competition Assigned
Sync Word	Competition Assigned

Model Rocket mission control software and ground systems

Electrical Connections & Power Distribution

PIN MAPPING (EXAMPLE — FINALIZE PER HARDWARE DESIGN)

ESP32 Pin	Device	Interface
GPIO21 / 22	IMU (SDA / SCL)	I²C
GPIO16 / 17	GPS (TX / RX)	UART
GPIO34	Battery Monitor	ADC
GPIO18 / 19	Radio (TX / RX)	UART

POWER DISTRIBUTION

Battery ⟶ Main Switch ⟶ Buck Converter

5V — GPS, USB Devices

3.3V — ESP32, IMU, Baro, XBee, 868 Radio

Device	Voltage	Current
ESP32	3.3 V	TBD
GPS	5 V	TBD
IMU / Barometer	3.3 V	TBD
XBee / 868 Radio	3.3 V	TBD

Model Rocket mission control software and ground systems

Sensor Integration

ESP32

I²C BUS

IMU

Orientation & acceleration

Barometer

Altitude measurement

UART

GPS

Position & time

Radio Module

Telemetry transmission

ADC

Battery Monitor

Battery voltage

Model Rocket mission control software and ground systems

Telemetry Packet Structure

Sensors → ESP32 → Packet Builder → Radio Module → Ground Station

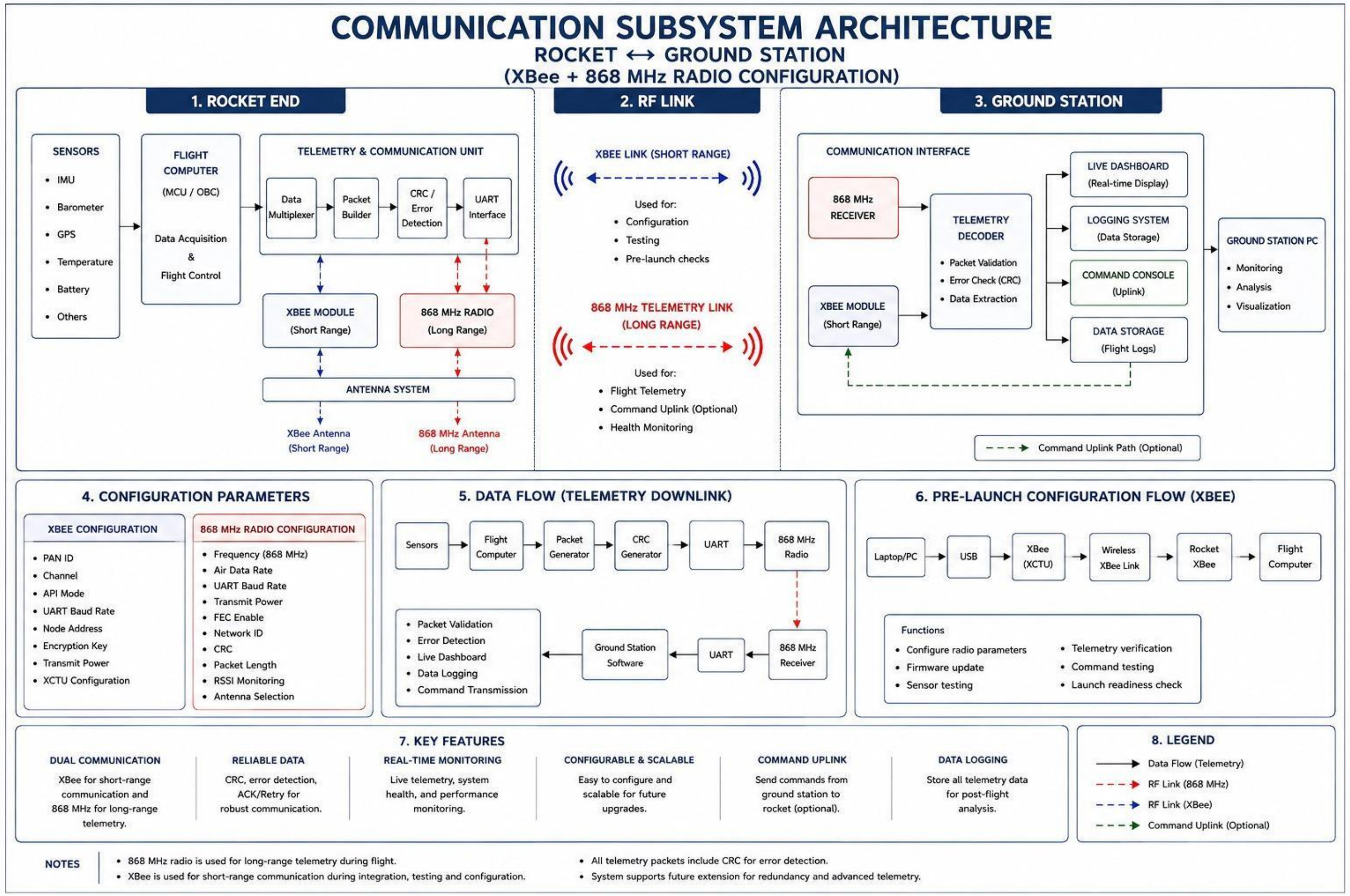
EXAMPLE TELEMETRY PACKET

Field	Description
Timestamp	Mission time
Packet ID	Sequence number
Flight State	Current mission phase
Altitude / GPS Lat/Lon	Position data
Battery Voltage	System health
RSSI	Communication quality
CRC	Error detection

PACKET VALIDATION

- Packet integrity check
- Packet ID sequence validation
- CRC verification (if implemented)
- Valid sensor range checking
- Communication status confirmation

Model Rocket mission control software and ground systems



Model Rocket mission control software and ground systems

Testing & Validation Plan & Risk Analysis

Hardware Verification

Sensors, ESP32, radios, power

4

Communication Verification

Radio config, RF link, range test

Sensor Integration

Simultaneous reads, accuracy, packet gen.

Ground Station Validation

Receive, display, log telemetry

End-to-End System Test

Full mission workflow simulation

Risk	Impact	Mitigation
Radio Interference	Telemetry loss	Backup communication channel
Packet Corruption	Incorrect telemetry	Packet validation and CRC
Sensor Failure	Missing measurements	Continuous sensor health monitoring
Battery Voltage Drop	System shutdown	Battery monitoring and voltage alerts
Serial Comm. Failure	Ground Station disconnect	Automatic reconnection mechanism

Model Rocket mission control software and ground systems

Key Design Decisions Since the PDR

The Ground Station architecture, RF commands, and telemetry requirements from the PDR template carry forward. The decisions below finalize the implementation approach.

Category	Decision	Rationale
Flight Controller	**ESP32**	Low cost, large ecosystem, easy Python integration
Ground Station Language	**Python**	Fast GUI, plotting, and serial development
GUI Framework	**PySide6**	Modern, professional widgets over Tkinter/PyQt
Serial Communication	**pySerial**	Cross-platform industry standard
Ground Station Radio	**Commercial USB Radio**	Plug-and-play; no GS firmware required
Packet Format	**ASCII CSV**	Simple, human-readable, easy to log
Ground Station Firmware	**None**	Operator interface is custom Python software

Model Rocket mission control software and ground systems

Communication Link Design

TRANSMITTER POWER LEVELS

Scenario	Tx Power	Rationale
Short-Range (Bench / Pad Test)	Reduced (TBD dBm)	Avoids receiver front-end saturation at close range
Long-Range (Flight)	Maximum (TBD dBm)	Maximizes link margin at apogee / horizon distance

Maximum transmit power is bounded by ISM-band / competition EIRP limits — the long-range setting must not exceed the assigned ceiling.

Error Correction & Detection

- CRC implemented for error detection — corrupted packets are discarded
- Evaluate forward error correction (e.g. Hamming, Reed–Solomon) vs. detect-and-discard
- If no FEC: compensate with redundant/repeated transmission rate

RECEIVER SENSITIVITY & LINK BUDGET

Tx Power

+ Tx Ant. Gain

− Path Loss (max range)

+ Rx Ant. Gain

≥ Rx Sensitivity + Fade Margin

Rx Sensitivity: TBD • Fade Margin Target: 10–20 dB

Modulation Techniques

Link	Modulation
2.4 GHz XBee (802.15.4)	O-QPSK (fixed by standard)
868 MHz Telemetry	LoRa (CSS) / FSK — TBD

Higher spreading factor / narrower bandwidth trades data rate and power draw for range.

Model Rocket mission control software and ground systems

Antenna Integration & Rotational Coverage

Antenna placement, quantity, and gain pattern must account for continuous rotational movement of the vehicle in flight.

Antenna Placement

- Mounted on the airframe exterior (nose cone / body tube), clear of obstruction
- RF-opaque materials (carbon fiber, metal fin cans) require external mounting or an RF-transparent window
- 868 MHz and 2.4 GHz antennas physically separated to minimize mutual coupling

Antenna Quantity & Diversity

- Single omnidirectional antenna per link as baseline; dual-antenna diversity as an option
- Diversity scheme selects/switches based on received signal strength (RSSI)
- Redundancy reduces impact of a single antenna null during flight

Gain Pattern vs. Rocket Rotation

- Roll and tumble sweep the antenna pattern relative to the ground station, risking periodic nulls
- Favor low-gain, broad/omnidirectional patterns (dipole, monopole, quadrifilar helix) over directional patches
- Circular polarization reduces orientation-dependent fading from spin vs. linear polarization

Link Budget Trade-off

- Lower antenna gain (better rotational coverage) reduces link margin
- Offset with higher transmit power or improved receiver sensitivity to preserve fade margin
- Ground station pattern (fixed omni / wide-beam / tracking) chosen to complement rocket-side coverage

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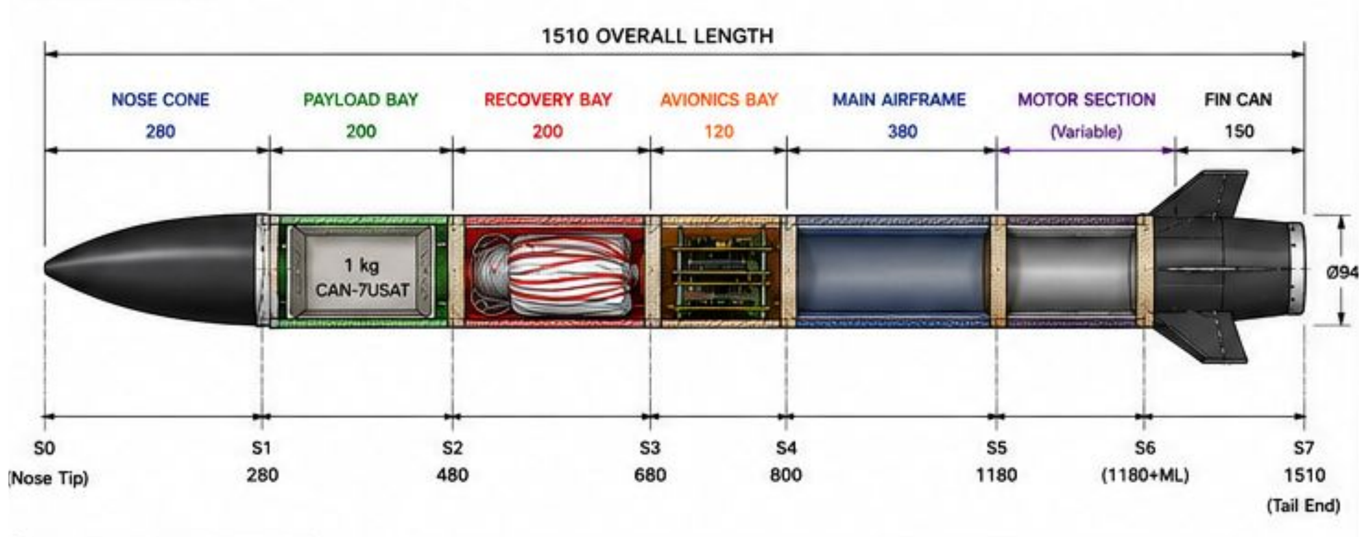

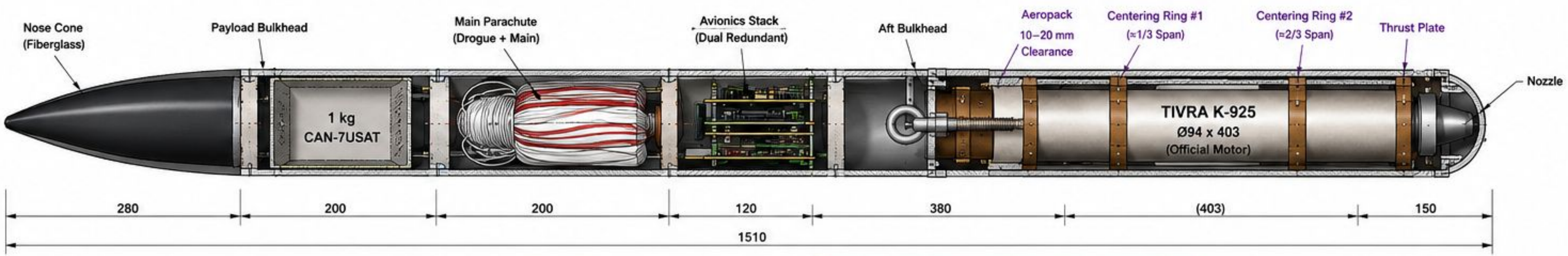

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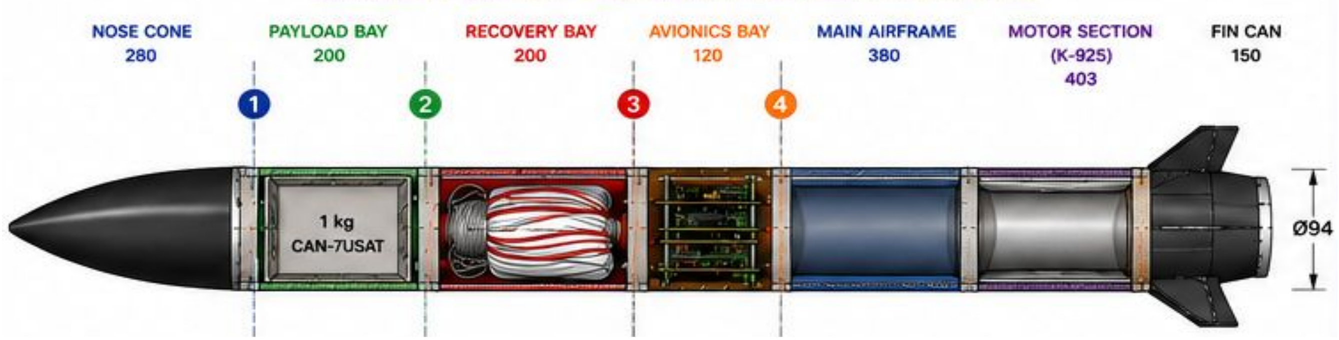
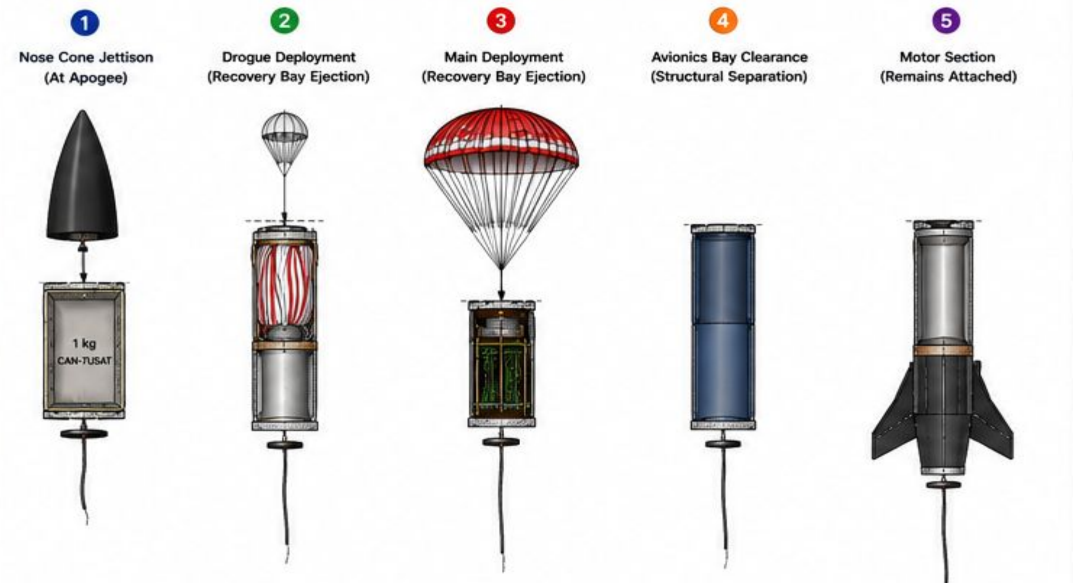
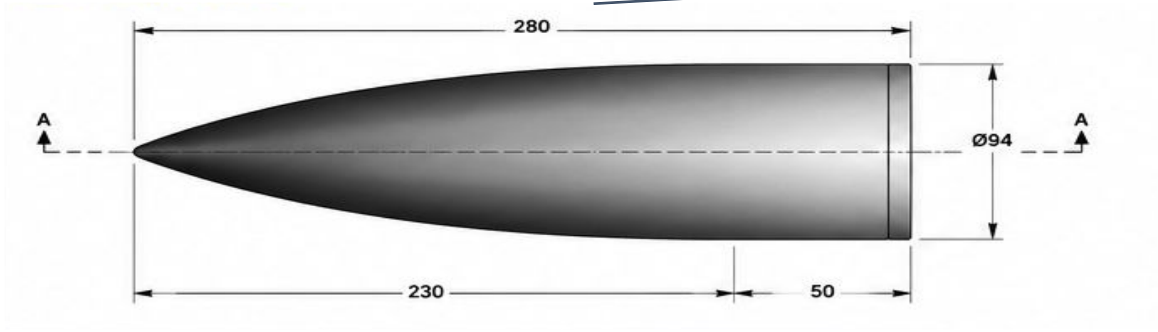


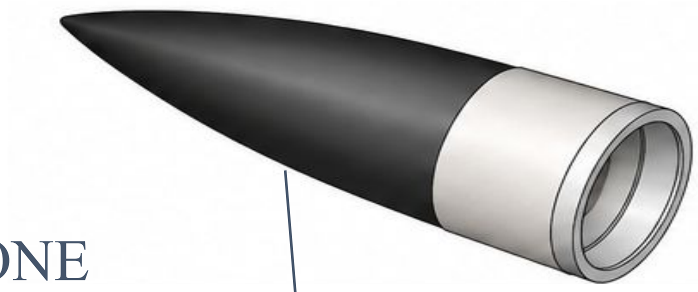
POINTS OF SEPERATION



SIDE VIEW

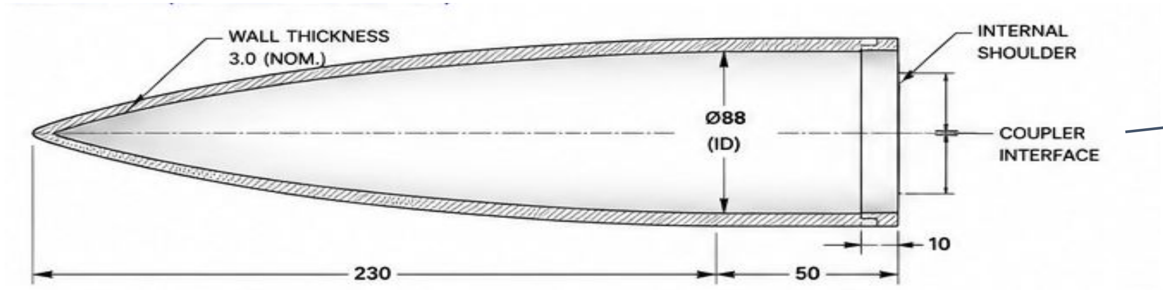


NOSE CONE
AF - 001



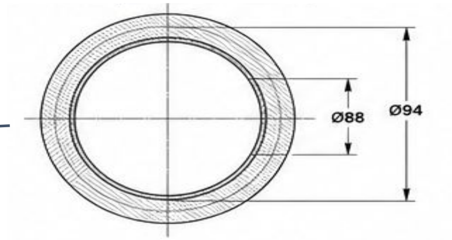
ISOMETRIC VIEW

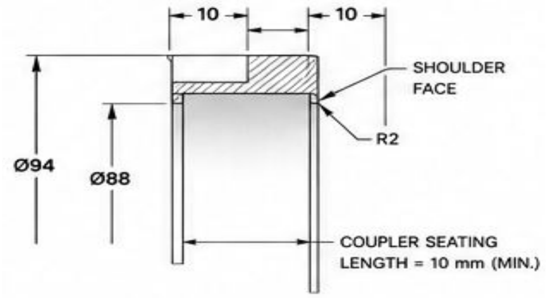
Material	Quantity
Fiberglass Composite	1



LONGITUDINAL SECTION VIEW

BASE VIEW





DIMENSIONS	
Overall Length	280 mm
Ogive Length (L)	230 mm
Base Length (Shoulder)	50 mm
Base Diameter (OD)	Ø94 mm
Wall Thickness	3.0 mm (Nom.)
Internal Diameter (ID)	Ø88 mm (Nom.)
Total Mass (Est.)	~ 420 g

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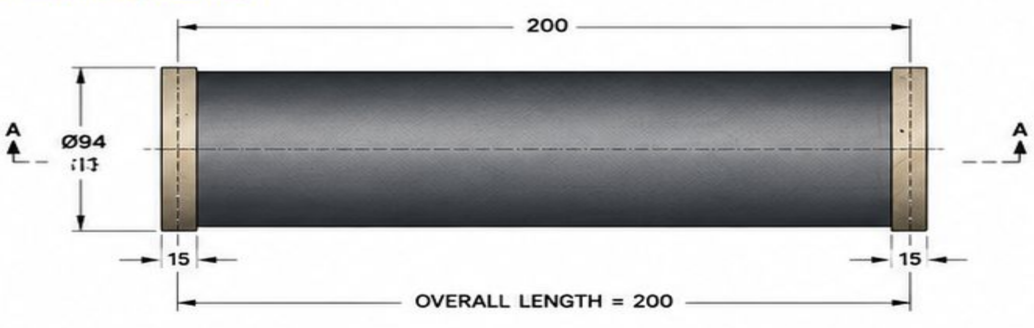
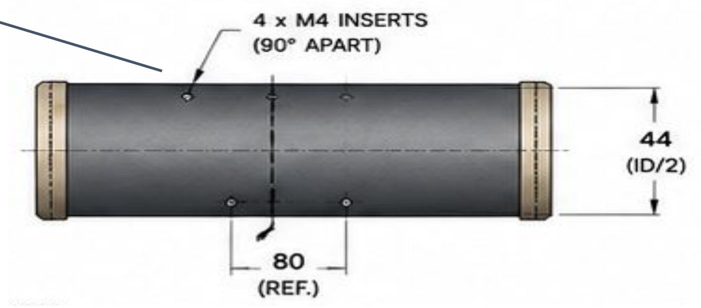


SIDE VIEW

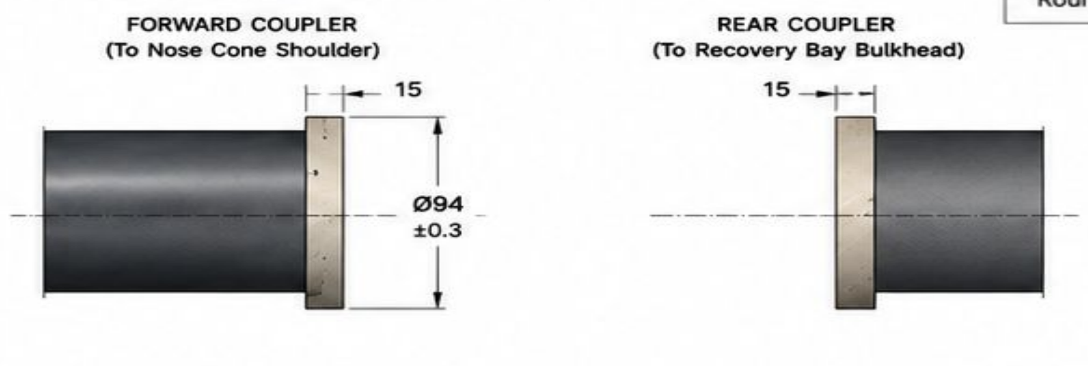
PAYLOAD MOUNTING HOLES



PAY LOAD BAY TUBE
AF - 002

ISOMETRIC VIEW

PAYLOAD BAY COUPLER INTERFACE



TOLERANCES	
Linear Dimensions	±0.5 mm
Diameter (OD, ID)	±0.3 mm
Wall Thickness	±0.3 mm
Perpendicularity	±0.5°
Roundness	±0.3 mm

Material	Fiberglass Composite
Resin	Epoxy
Layup Schedule	(±45 / 0)2S
Cure Method	Room Temp. / Post Cure Recommended
Recommended Pressure	Vacuum Bagging
Min. Cure Temp.	25 °C
Min. Full Cure Time	24 hrs

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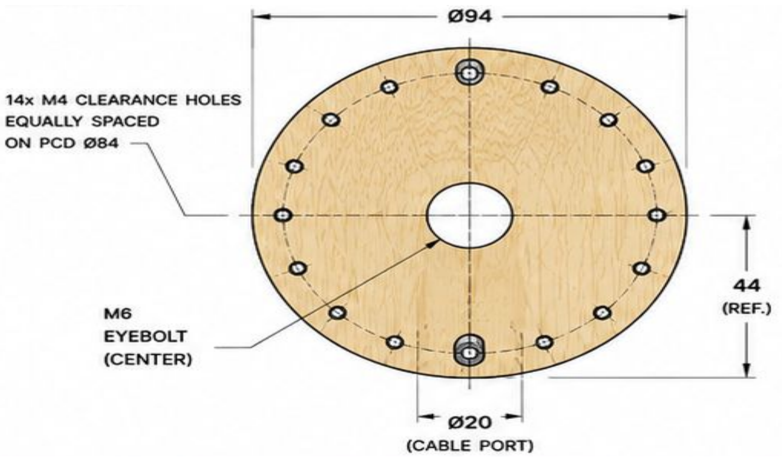
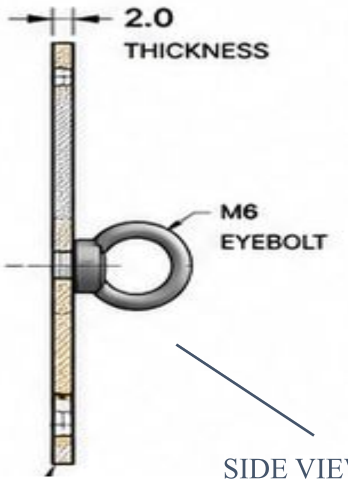
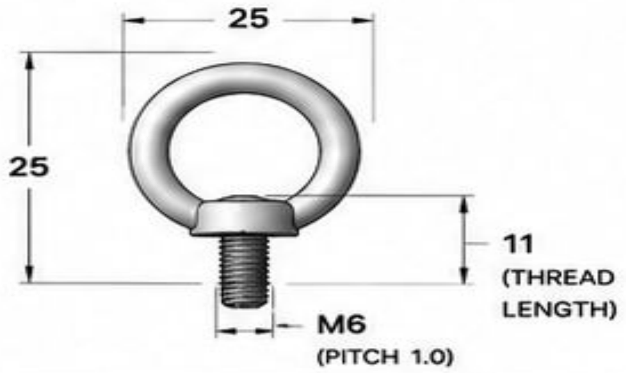
PAY LOAD BULK HEAD
AF - 003



FRONT VIEW

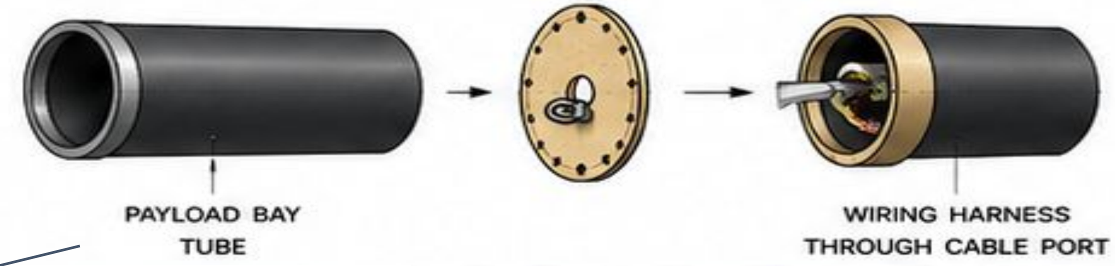


SIDE VIEW



EVE BOLT

DIMENSIONS	
Outer Diameter (OD)	Ø94 mm
Thickness	2.0 mm
Center Hole (Cable Port)	Ø20 mm
Eyebolt	M6 (Center)
Bolt Circle (PCD)	Ø84 mm
No. of Holes	14 (M4 Clearance)
Hole Size	Ø4.5 mm
Mass (Est.)	~ 22 g



INSTALLATION REFERENCE

ITEM	DESCRIPTION	MATERIAL	QTY
1	Payload Bulkhead	Birch Plywood (2.0 mm)	1
2	Eyebolt M6	Stainless Steel	1
3	Nut M6	Stainless Steel	1

TOLERANCES	
Linear Dimensions	±0.3 mm
Hole Diameter	±0.1 mm
PCD	±0.2 mm
Thickness	±0.2 mm
Flatness	±0.2 mm

Team ID :
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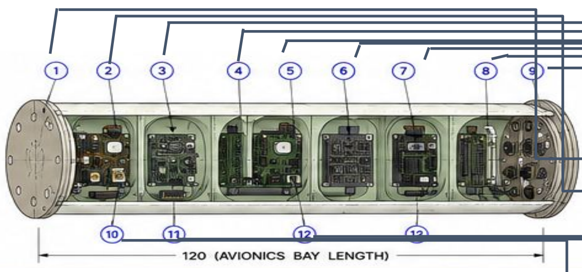
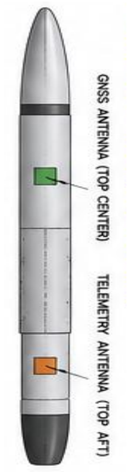
AVIONICS BAY
AF - 004

INTERNAL LAYOUT



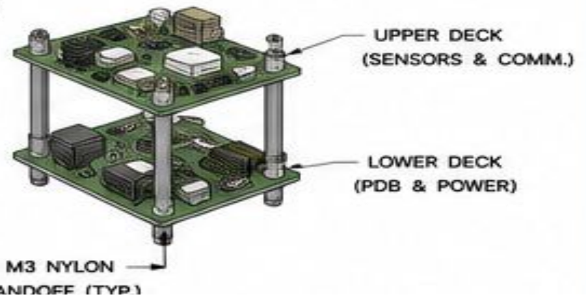
NO.	COMPONENT	FUNCTION
1	Power Distribution Board (PDB)	Power Management
2	Flight Controller (MCU)	Primary Control & Logging
3	IMU (LSM6DS3 + BMI088)	Attitude & Motion Sensing
4	Barometric Altimeter (BMP388)	Altitude Measurement
5	GNSS Module (NavIC / GPS)	Position & Time
6	Telemetry Transceiver (LoRa)	RF Telemetry Uplink
7	Data Logger (SD Card Module)	Data Storage
8	Backup Altimeter (CAN-7USAT)	Redundant Altitude
9	Pyro Switch Board	Deployment Control
10	Voltage Monitor (INA219) + ADC (ADS1115)	Power & Sensor Monitoring
11	Watchdog Timer	System Health Monitoring
12	Buzzer / Beacon Driver	Recovery Indication
13	Connectors (Bulkhead Pass-Through)	Interface to Other Bays



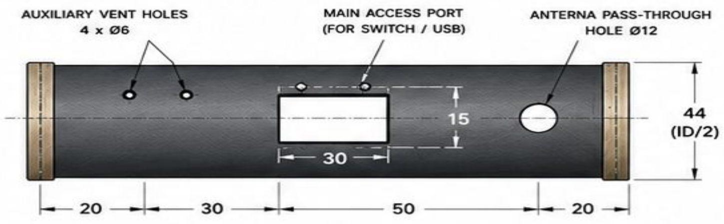
ANTENNA ALLOCATION

MOUNTING AND STANDOFFS

- All boards mounted on a 2-level stack using M3 nylon standoffs (8 mm height).
- All sensors vibration isolated using foam pads.
- All connectors are locking type.
- All wiring secured using cable tie mounts.
- Avionics bay closed with O-ring sealed bulkheads.



ACCESS AND MOUNTING FEATURES

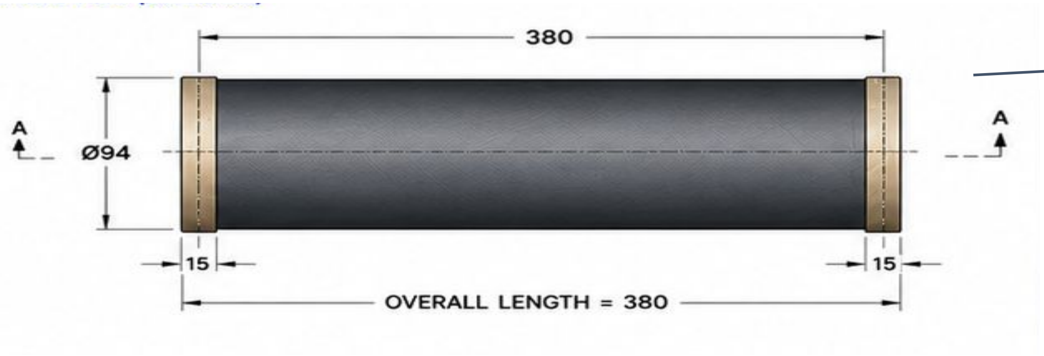


DIMENSIONS	
Overall Length	120 mm
Outer Diameter (OD)	Ø94 mm
Inner Diameter (ID)	Ø88 mm
Wall Thickness	3.0 mm (Nom.)
Material	Fiberglass Composite
Mass (Est.)	~ 110 g

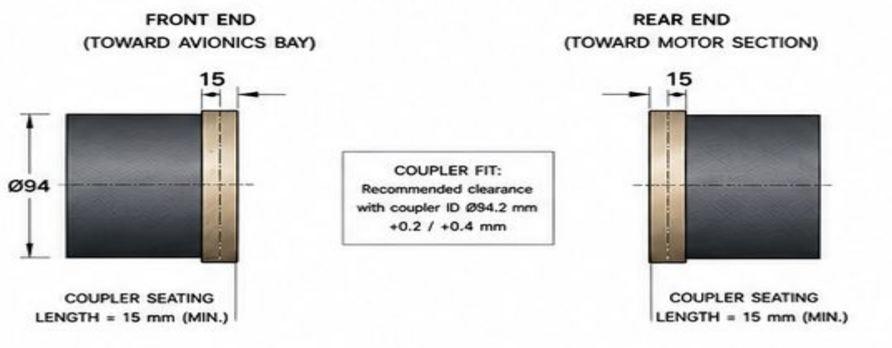
TOLERANCES	
Linear Dimensions	±0.5 mm
Diameter (OD, ID)	±0.3 mm
Wall Thickness	±0.3 mm
Perpendicularity	±0.5°
Roundness	±0.3 mm

Team ID :
2026-INSPACe-ROCKETRY-059

SIDE VIEW

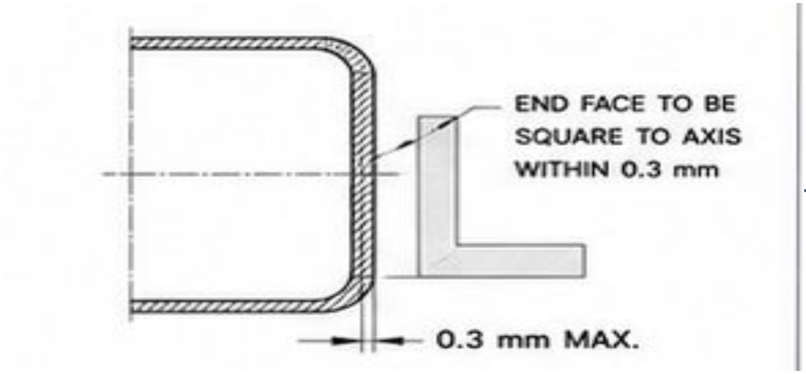


COUPLER INTERFACE



MAIN AIRFRAME
AF - 005

END SQUARENESS



ITEM	DESCRIPTION	MATERIAL	QTY
1	Main Airframe Tube	Fiberglass Composite	1

DIMENSIONS	
Overall Length	380 mm
Outer Diameter (OD)	Ø94 mm
Inner Diameter (ID)	Ø88 mm
Wall Thickness	3.0 mm (Nom.)
Material	Fiberglass Composite
Mass (Est.)	~ 340 g

TOLERANCES	
Linear Dimensions	±0.5 mm
Diameter (OD, ID)	±0.3 mm
Wall Thickness	±0.3 mm
Perpendicularity	±0.5°
Roundness	±0.3 mm

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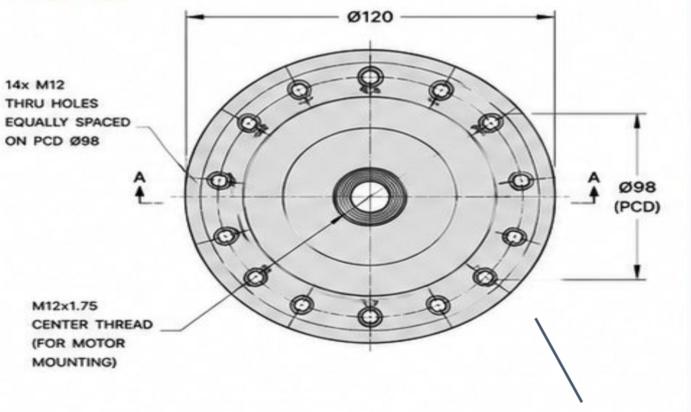
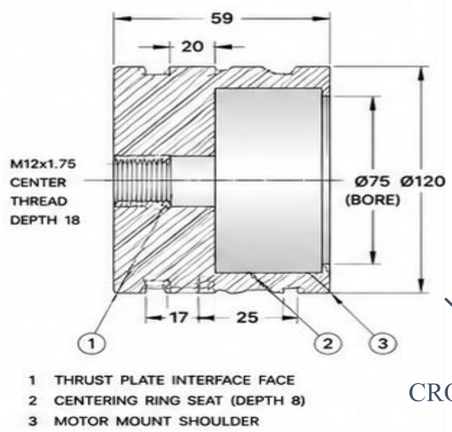
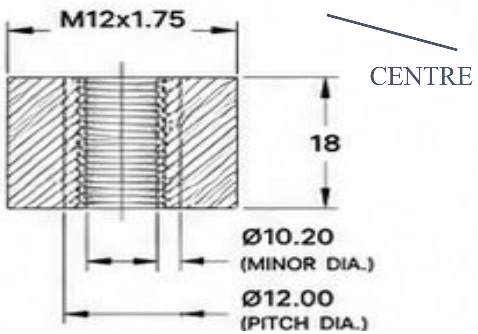


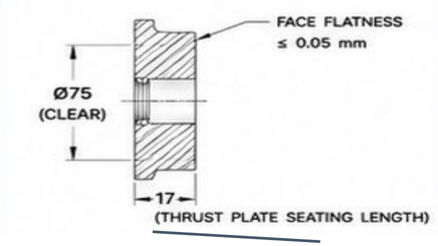
FRONT VIEW



CROSS SECTIONAL VIEW

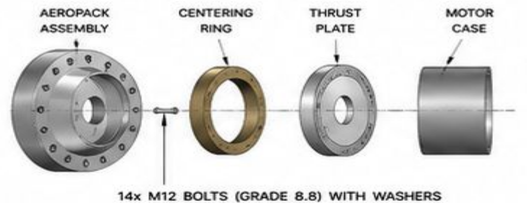


CENTRE THREAD



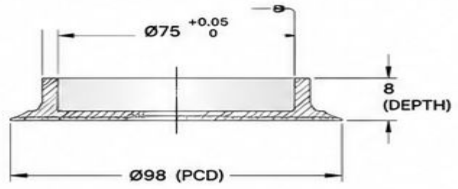
THRUST PLATE INTERFACE

MOUNTING REFERENCE



AEROPACK ASSEMBLY
AF - 006

ITEM	DESCRIPTION	MATERIAL	QTY
1	Aeropack Body	Aluminum 6061-T6	1
2	Centering Ring (2026 Spec)	Phenolic Composite	1
3	Thrust Plate (2026 Spec)	Aluminum 7075-T6	1
4	Hex Bolt M12x35 (Grade 8.8)	Steel (Zn Plated)	14
5	Flat Washer M12	SS 304	14
6	Spring Washer M12	SS 304	14

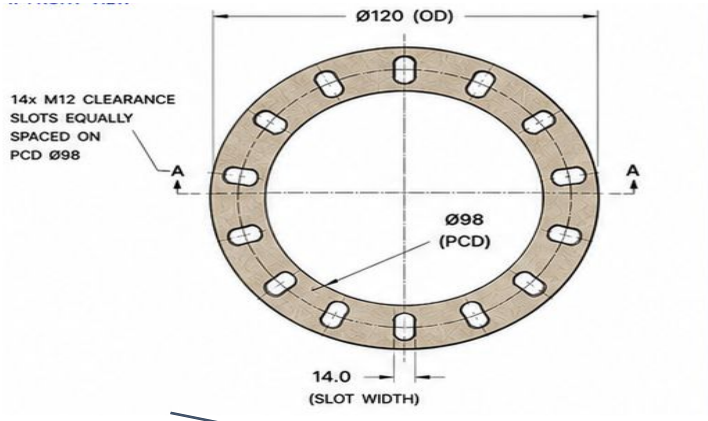


CENTERING RING SEAT

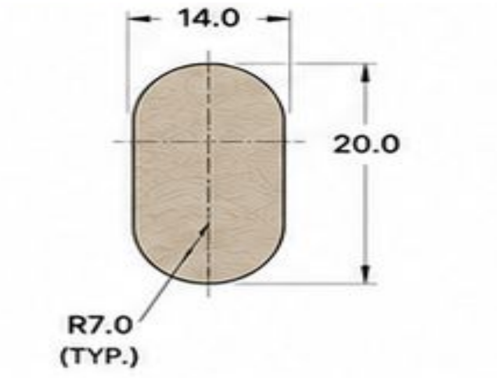
DIMENSIONS	
Overall Diameter	Ø120 mm
Overall Length	59 mm
Bolt Circle Diameter (PCD)	Ø98 mm
Number of Bolt Holes	14
Bolt Hole Size	Ø13.0 mm (for M12)
Center Thread	M12x1.75
Thread Depth	18 mm
Bore Diameter (Clear)	Ø75 mm
Material	Aluminum 6061-T6
Mass (Est.)	~ 620 g

TOLERANCES	
Linear Dimensions	±0.2 mm
Diameters	±0.1 mm
Bolt Hole Position (PCD)	±0.15 mm
Perpendicularity	0.03 mm
Runout (Face to Bore)	0.05 mm

MODEL ROCKETRY INDIA



FRONT VIEW



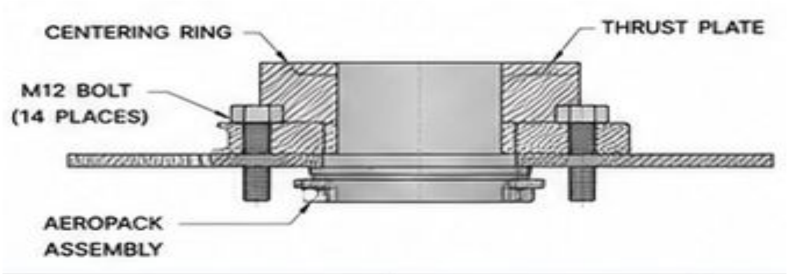
BOLT SLOT DETAIL

ITEM	DESCRIPTION	MATERIAL	QTY
1	Centering Ring	Phenolic Composite	1

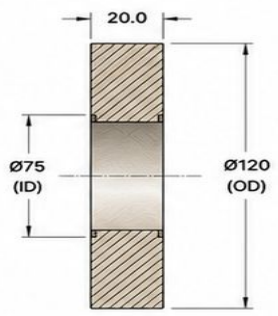
CENTERING RING
AF - 007

DIMENSIONS	
Outer Diameter (OD)	Ø120 mm
Inner Diameter (ID)	Ø75 mm
Thickness	20.0 mm
Bolt Circle Diameter (PCD)	Ø98 mm
No. of Bolt Slots	14
Slot Width	14.0 mm
Slot Length (Radial)	20.0 mm
Slot Type	Straight
Material	Phenolic Composite
Mass (Est.)	~ 70 g

TOLERANCES	
Linear Dimensions	±0.2 mm
Diameters	±0.1 mm
Slot Width	±0.1 mm
Slot Position (PCD)	±0.15 mm
Perpendicularity	0.03 mm
Parallelism	0.03 mm
Flatness	0.03 mm

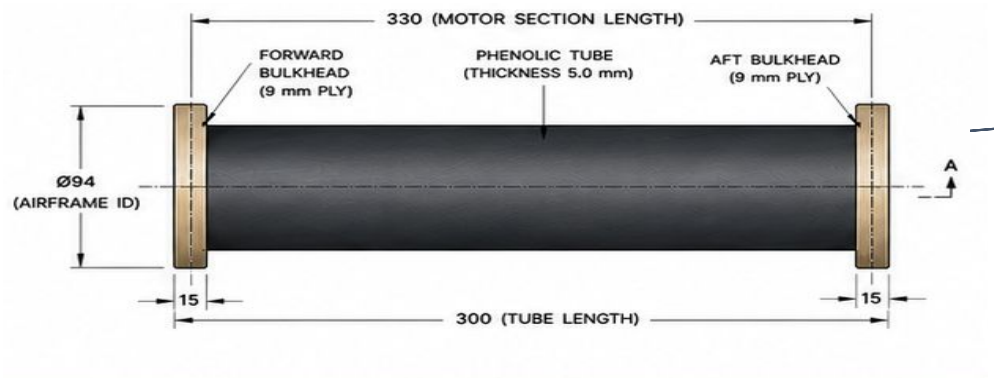


MOUNTING REFERENCE



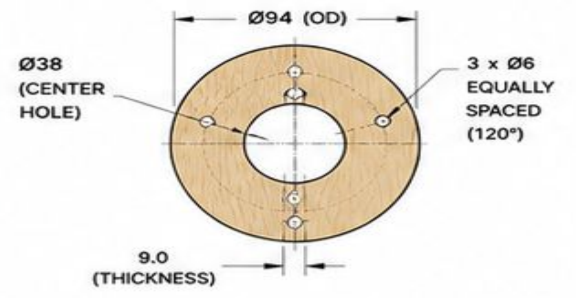
SECTIONAL AND ISOMETRIC VIEW

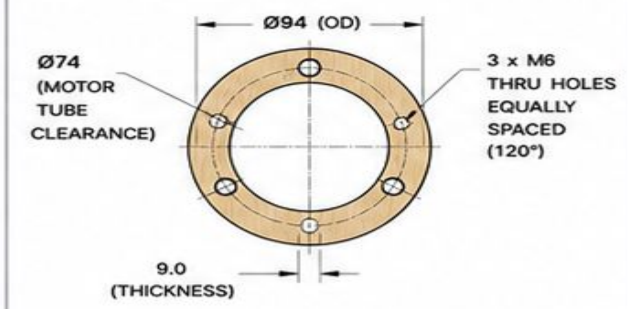
Team ID :
2026-INSPACe-ROCKETRY-059



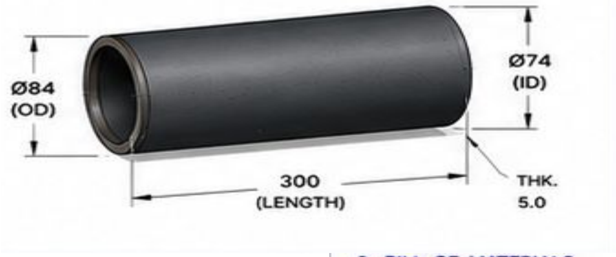
SIDE VIEW

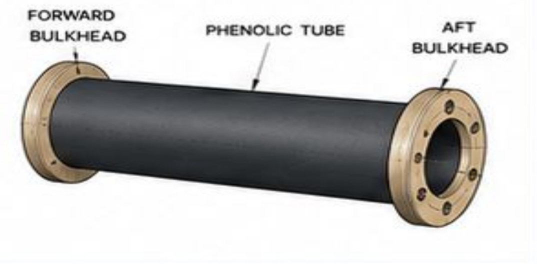
FORWARD BULKHEAD

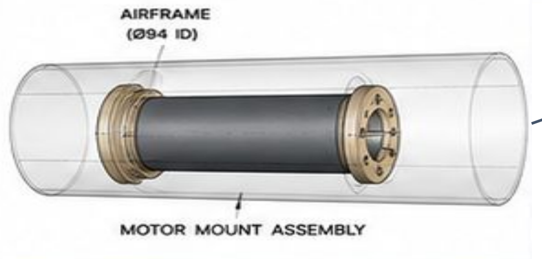




MOTOR MOUNT ASSEMBLY
AF - 008







ASSEMBLY REFERENCE

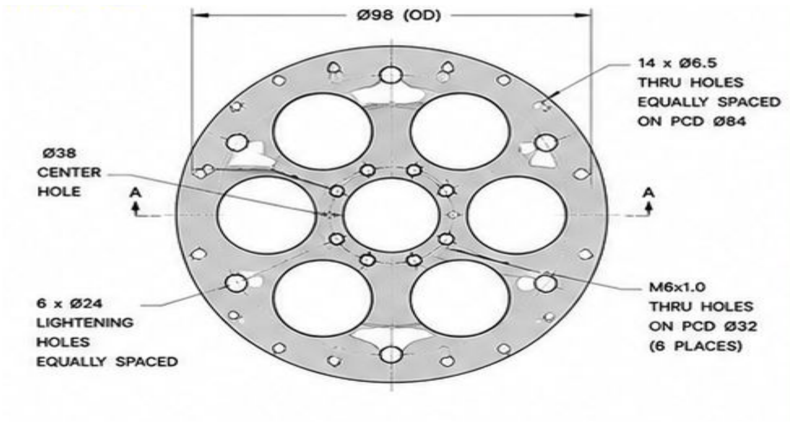
9. BILL OF MATERIALS

ITEM	DESCRIPTION	MATERIAL	QTY
1	Phenolic Tube	Phenolic Composite (Ø84 x ID Ø74 x L300)	1
2	Forward Bulkhead	Birch Plywood (9 mm)	1
3	Aft Bulkhead	Birch Plywood (9 mm)	1
4	Socket Head Bolt M6 x 20	SS 304	3
5	Flat Washer M6	SS 304	3
6	Epoxy Adhesive	Structural Epoxy	-

TOLERANCES	
Linear Dimensions	±0.3 mm
Diameters	±0.2 mm
Coaxiality (Tube to ID)	±0.3 mm
Squareness (Bulkhead)	±0.3 mm
Perpendicularity	±0.3 mm
Runout (Max.)	0.05 mm

DIMENSIONS	
Motor Section Length	330 mm
Airframe Inner Diameter	Ø94 mm
Tube Outer Diameter (OD)	Ø84 mm
Tube Inner Diameter (ID)	Ø74 mm
Tube Thickness	5.0 mm
Bulkhead Thickness	9.0 mm
Bulkhead Material	Birch Plywood
Tube Material	Phenolic Composite
Mass (Est.)	~ 210 g

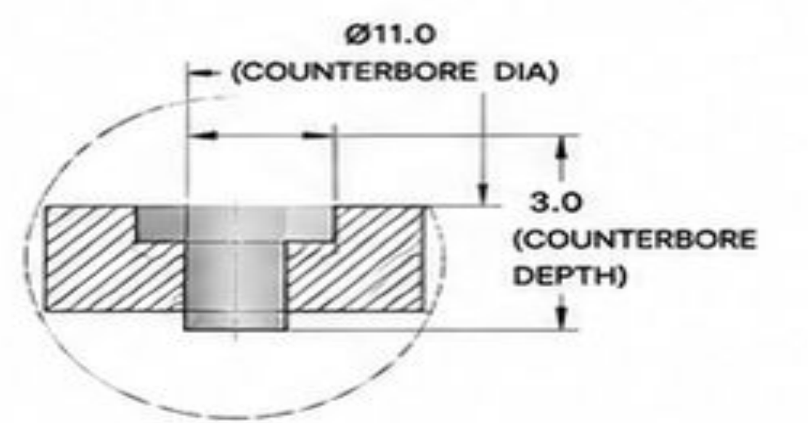
Team ID :
2026-INSPACe-ROCKETRY-059



ITEM	DESCRIPTION	MATERIAL	QTY	REMARKS
1	Thrust Plate	Aluminum 7075-T6	1	-

MATERIAL PROPERTIES (7075-T6)	
Density	2.81 g/cm³
Tensile Strength	≥ 572 MPa
Yield Strength	≥ 503 MPa
Hardness	150 HB
Modulus of Elasticity	71.7 GPa

THRUST PLATE
AF - 009

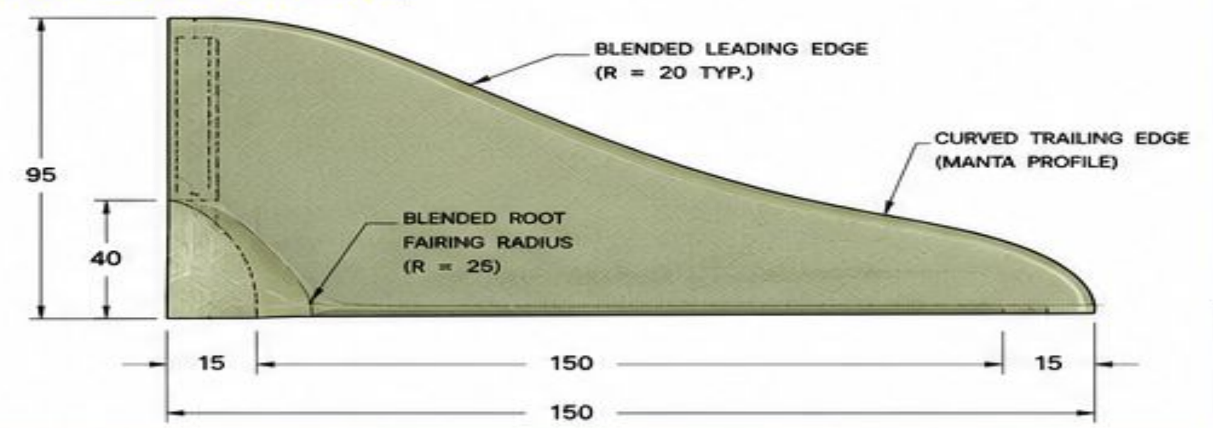


DIMENSIONS	
Outer Diameter (OD)	Ø98 mm
Thickness	8.0 mm
Center Hole Diameter	Ø38 mm
Bolt Circle Diameter (PCD-1)	Ø84 mm (14 Holes)
Bolt Circle Diameter (PCD-2)	Ø32 mm (6 Holes)
Bolt Hole Size (Outer)	Ø6.5 mm (14 Holes)
Bolt Hole Size (Inner)	M6x1.0 (6 Holes)
Lightening Hole Diameter	Ø24 mm (6 Holes)
Material	Aluminum 7075-T6
Mass (Est.)	~ 120 g

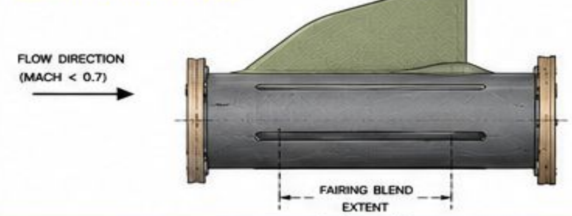
TOLERANCES	
Linear Dimensions	±0.1 mm
Diameters	±0.1 mm
Hole Position (PCD)	±0.1 mm
Perpendicularity	0.03 mm
Flatness	0.05 mm
Surface Finish (Ra)	≤ 1.6 µm

Team ID :
2026-INSPACe-ROCKETRY-059

BFMS FIN
AF - 010

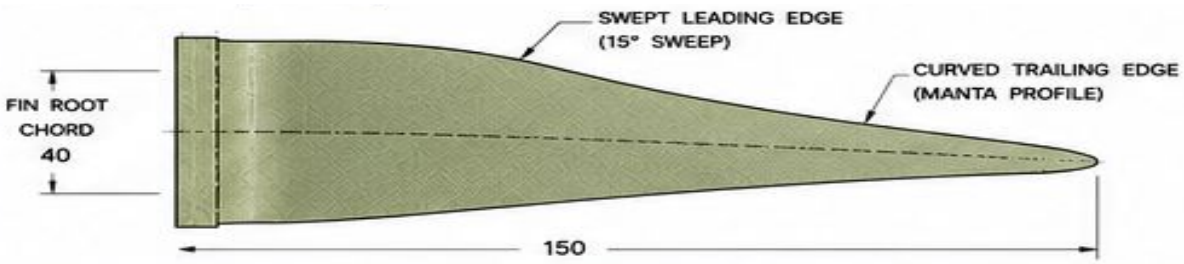


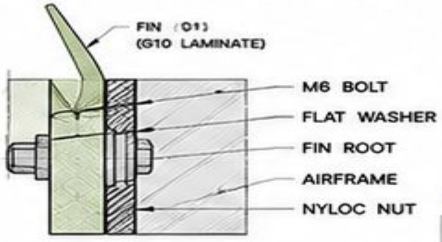
2. SIDE VIEW (FIN ATTACHED)



DIMENSIONS	
Fin Span (Height)	95 mm
Fin Root Chord	40 mm
Fin Tip Chord	15 mm
Fin Sweep Length	150 mm
Fin Thickness	3.0 mm
Leading Edge Radius	R3 mm
Trailing Edge Radius	R1 mm
Root Edge Thickness	6.0 mm
Fin Area (One Fin)	~ 59.3 cm²
Number of Fins	4 (BFMS)
Material	G10 Fiberglass Laminate
Mass (Est.)	~ 45 g (One Fin)

TOLERANCES	
Linear Dimensions	±0.3 mm
Thickness	±0.15 mm
Hole Diameter	±0.1 mm
Hole Position (PCD)	±0.15 mm
Perpendicularity	±0.3°
Angular Tolerance	±0.5°
Surface Finish (Ra)	≤ 1.6 µm



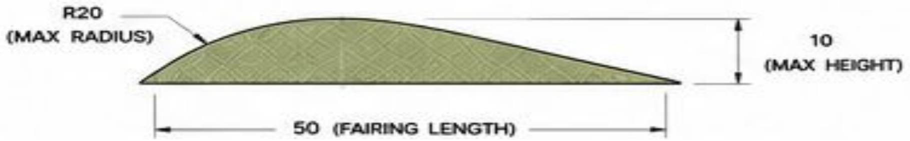


(TYPICAL FOR 4 PLACES)

10. BILL OF MATERIALS (ONE FIN)

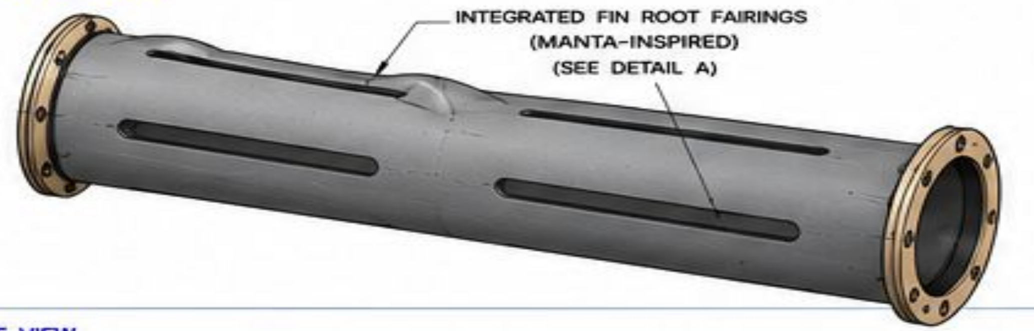
ITEM	DESCRIPTION	MATERIAL	QTY	REMARKS
1	Fin (BFMS Type)	G10 Fiberglass Laminate	1	-
2	Socket Head Bolt M6 x 20	SS 304	4	-
3	Flat Washer M6	SS 304	4	-
4	Nylock Nut M6	SS 304	4	-

MATERIAL PROPERTIES – G10 FIBERGLASS LAMINATE	
Density	1.90 g/cm³
Tensile Strength	≥ 340 MPa
Flexural Strength	≥ 400 MPa
Compressive Strength	≥ 300 MPa
Shear Strength	≥ 50 MPa



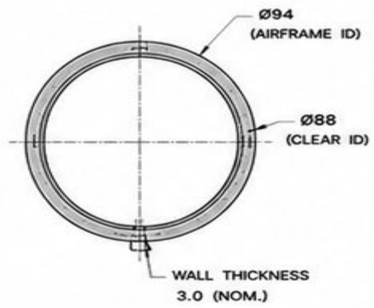
Team ID :
2026-INSPACe-ROCKETRY-059

. ISOMETRIC VIEW



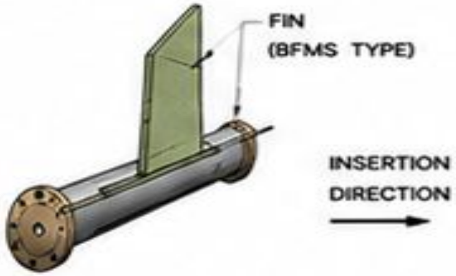
SIDE VIEW

380 (OVERALL LENGTH)



DIMENSIONS	
Overall Length	380 mm
Fin Can Length (Usable)	340 mm
Outer Diameter (OD)	Ø94 mm
Inner Diameter (ID)	Ø88 mm
Wall Thickness	3.0 mm (Nom.)
Slot Length	220 mm
Slot Width	8.0 mm
Number of Slots	4 (Equally Spaced)
End Flange Thickness	5.0 mm
Flange Outer Diameter	Ø84 mm
Flange PCD	Ø84 mm
Mounting Holes	6 x Ø6.5 mm
Material	Fiberglass Composite
Mass (Est.)	~ 120 g

TOLERANCES	
Linear Dimensions	±0.3 mm
Diameters	±0.2 mm
Hole Position (PCD)	±0.15 mm
Perpendicularity	±0.3 mm
Concentricity	±0.3 mm
Surface Finish (Ra)	≤ 1.6 µm



BFMS FIN CAN
AF - 011

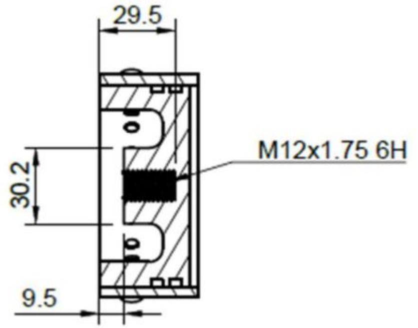
11. BILL OF MATERIALS

ITEM	DESCRIPTION	MATERIAL	QTY	REMARKS
1	Fin Can Body	Fiberglass Composite	1	-
2	Front End Flange	Fiberglass Composite	1	Bonded
3	Rear End Flange	Fiberglass Composite	1	Bonded
4	Socket Head Bolt M6 x 20	SS 304	12	-
5	Flat Washer M6	SS 304	12	-
6	Nyloc Nut M6	SS 304	12	-

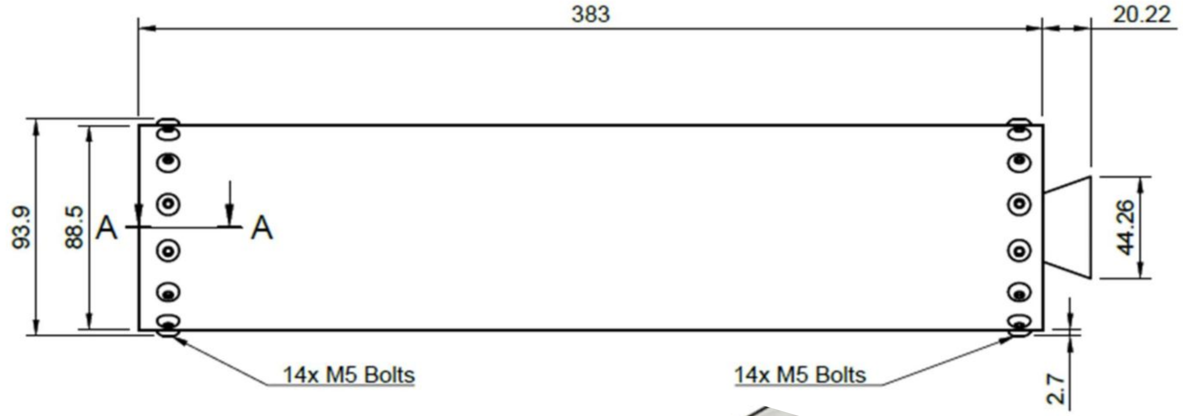
Team ID :
2026-INSPACe-ROCKETRY-059

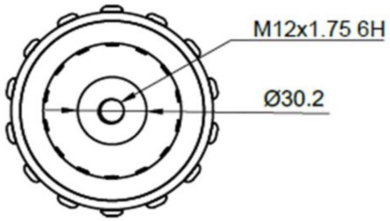
Units	Scale
mm	1:2

Motor : TIVRA K-925



Top
Interface





Name: **TIVRA K-925**

Motor Diameter: **94 MM**

Casing Length (Including Nozzle Overhang): **403 MM**

Casing Material: **Aluminium**

Pressure Rating: **12 MPa**

Total Impulse: **2500 Newtons-Sec**

Average Thrust: **704 Newtons**

Peak Thrust: **1559 Newtons**

Thrust Duration: **3.6 Sec**

Propellent Weight: **2426 Grams**

Motor Weight: **4459 Grams**

Total Liftoff Weight: **15000 Grams**

Mass & CG Location (From Top)		
	Mass	CG
Launch	4459 g	201.03 mm
Empty	2033 g	214.20 mm

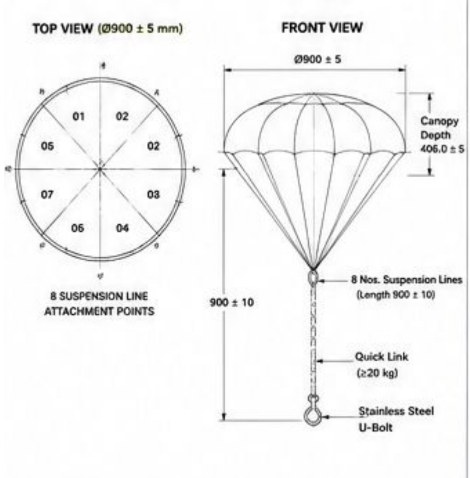
Team ID :
2026-INSPACe-ROCKETRY-059

Recovery Shock Load

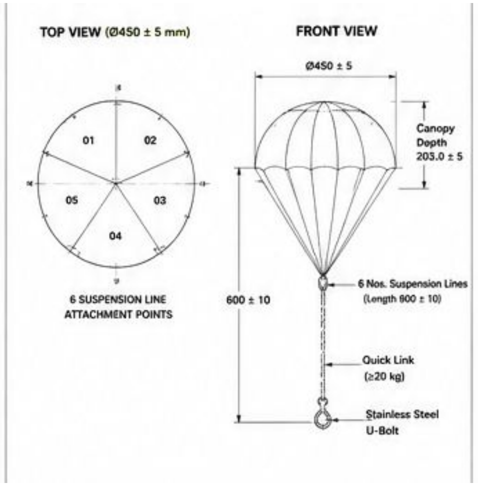
- Total lift-off mass (m) = 6.25 kg
- Peak recovery shock load (initial) = 24.5 kN
- Design load (SF = 2.0) = 49.0 kN
- Load path verified
- Structural components adequate for design load

DESCRIPTION	MASS (kg)
Rocket (without recovery system)	5.00
Recovery system (from document)	1.25
TOTAL LIFT-OFF MASS (m)	6.25 kg

Main Parachute



Drogue Parachute



Recovery System

ITEM	SPECIFICATION / VALUE
Recovery system mass (Total)	1.25 kg
Main parachute diameter	Ø900 ± 5 mm
Main parachute canopy depth	406.0 ± 5 mm
Main suspension lines	8 Nos.
Main line length	900 ± 10 mm
Drogue parachute diameter	Ø450 ± 5 mm
Drogue canopy depth	203.0 ± 5 mm
Drogue suspension lines	6 Nos.
Drogue line length	600 ± 10 mm
Shock cord (Kevlar) free length	100 mm
Quick link / Soft link	Rating ≥ 20 kg

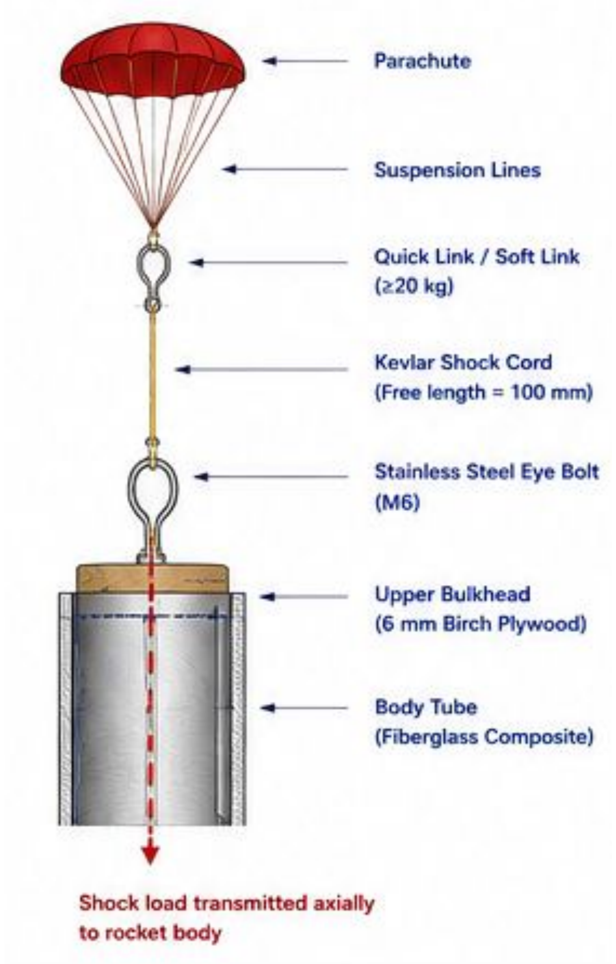
7.2 PEAK SHOCK LOAD (INITIAL)

Using $F_{shock} = \frac{m v^2}{2 s}$

m = 6.25 kg, v = 28 m/s, s = 0.10 m

$$F_{shock} = \frac{6.25 \times 28^2}{2 \times 0.10}$$

$$F_{shock} = 24{,}500 \text{ N} = 24.5 \text{ kN}$$

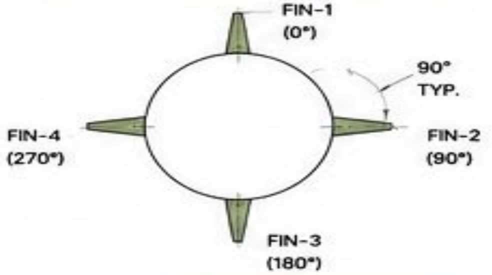


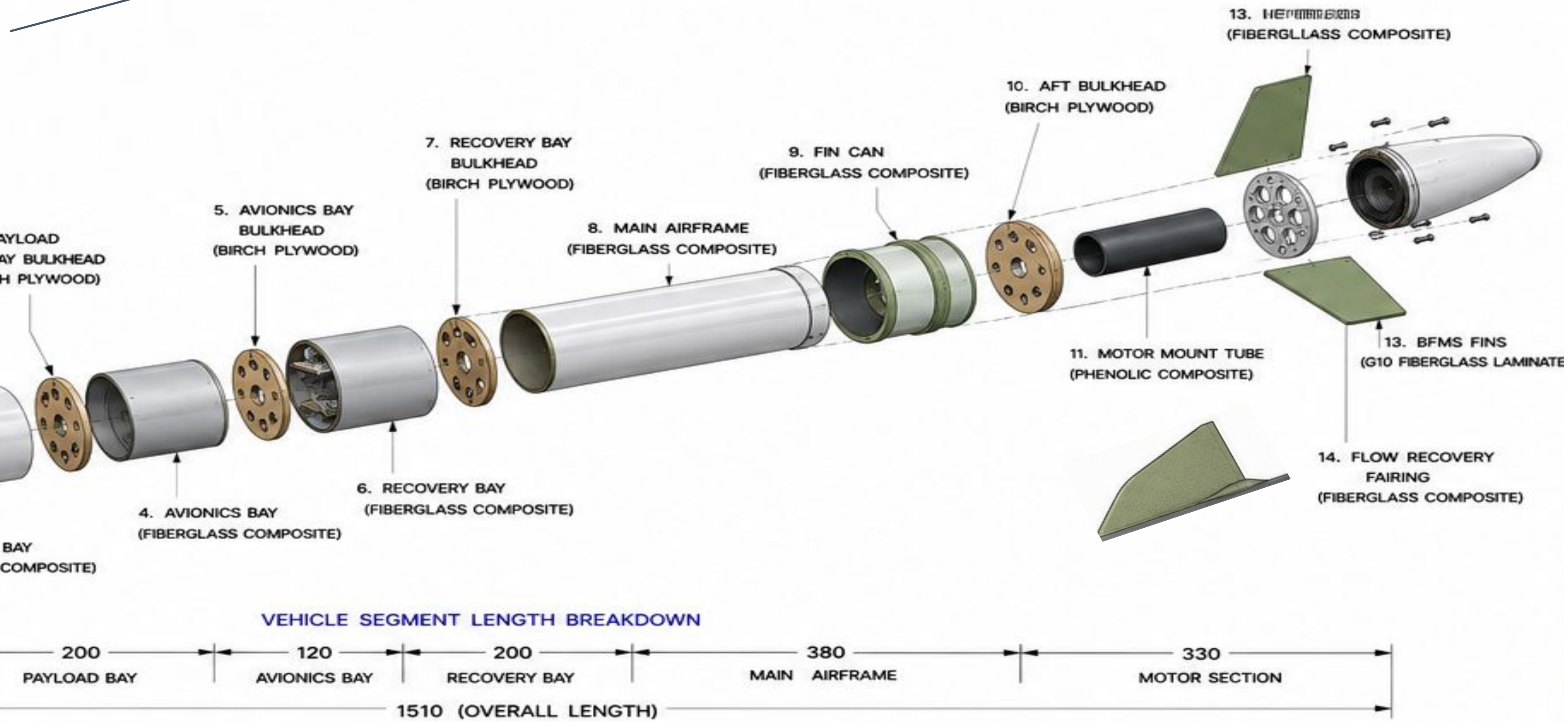
MODEL ROCKETRY INDIA

Team ID :
2026-INSPACe-ROCKETRY-059

EXPLODED VIEW
AF - 012

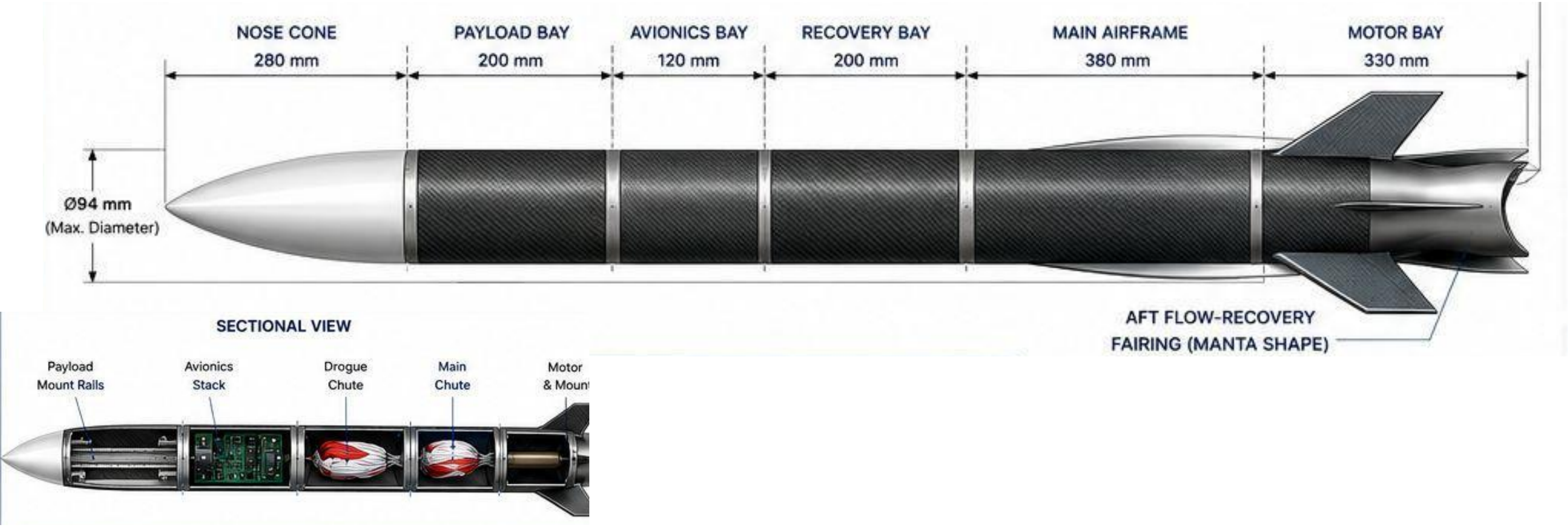
FIN SET ORIENTATION



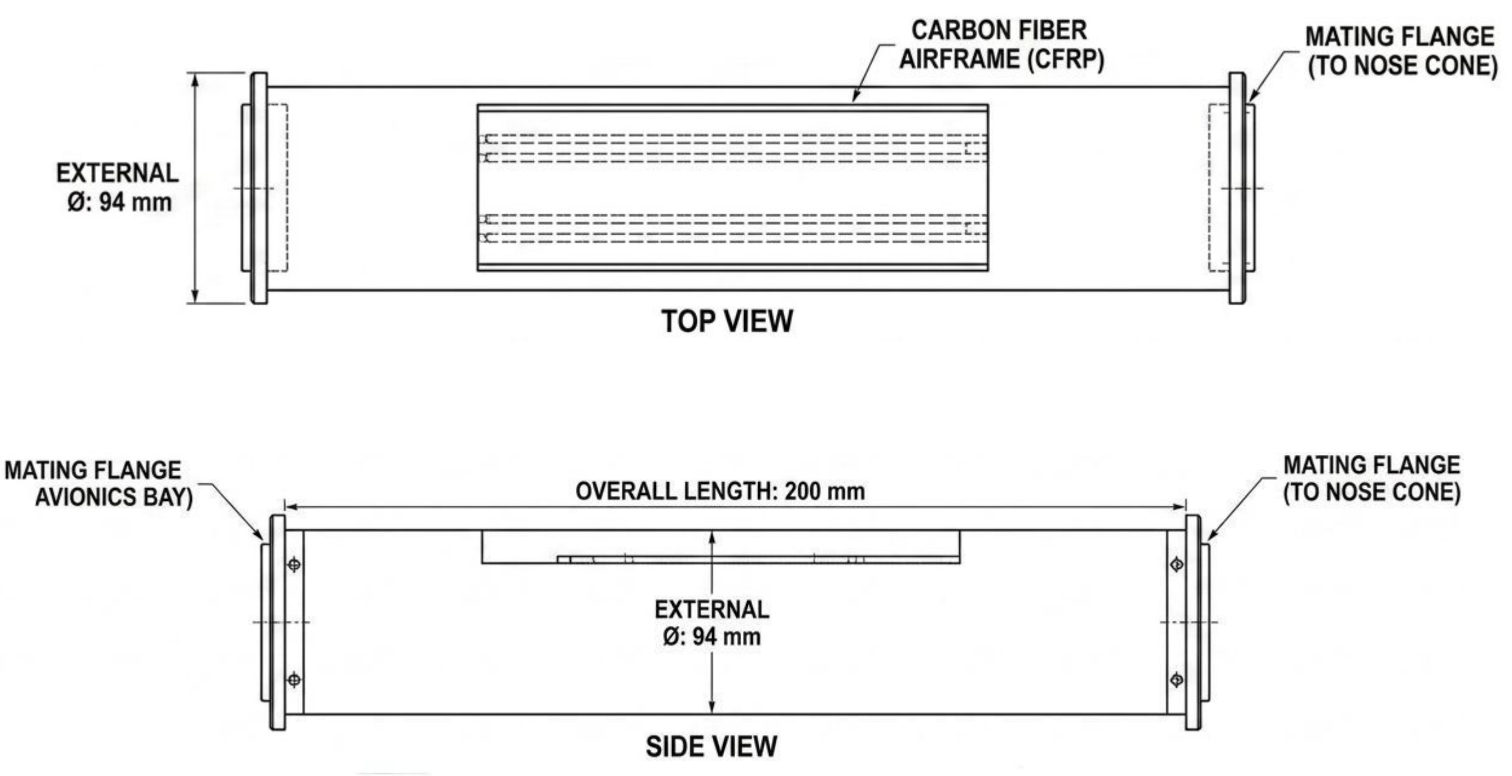


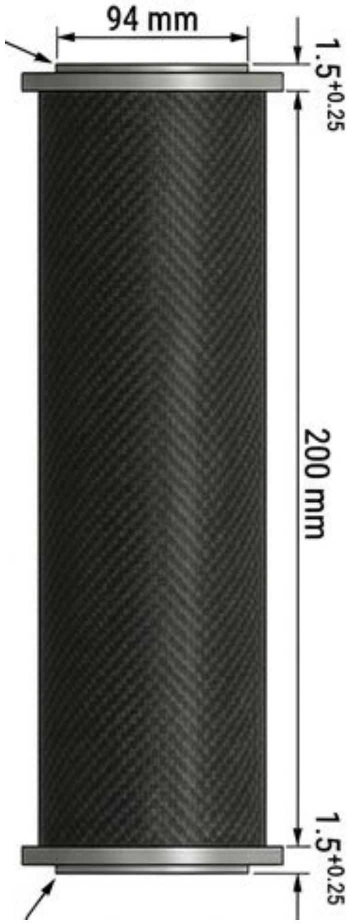
Payload System

Rocket configuration overview

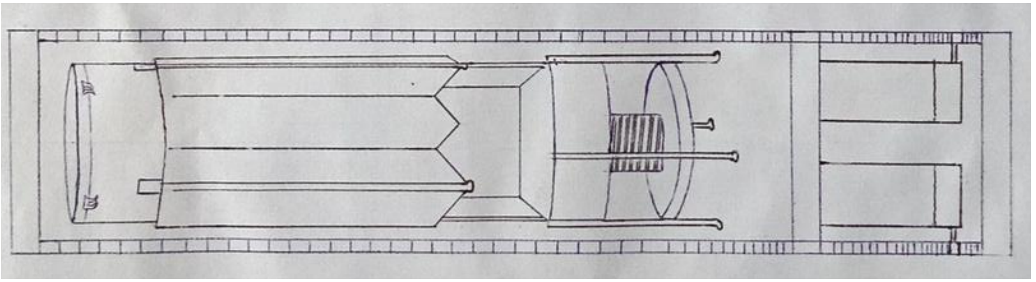


Payload Bay Structural Configuration

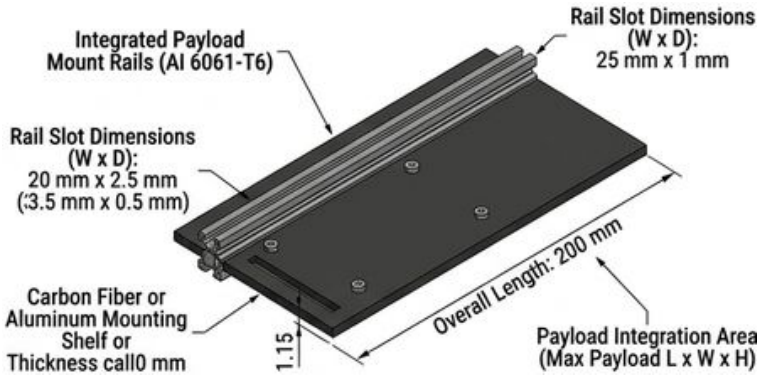


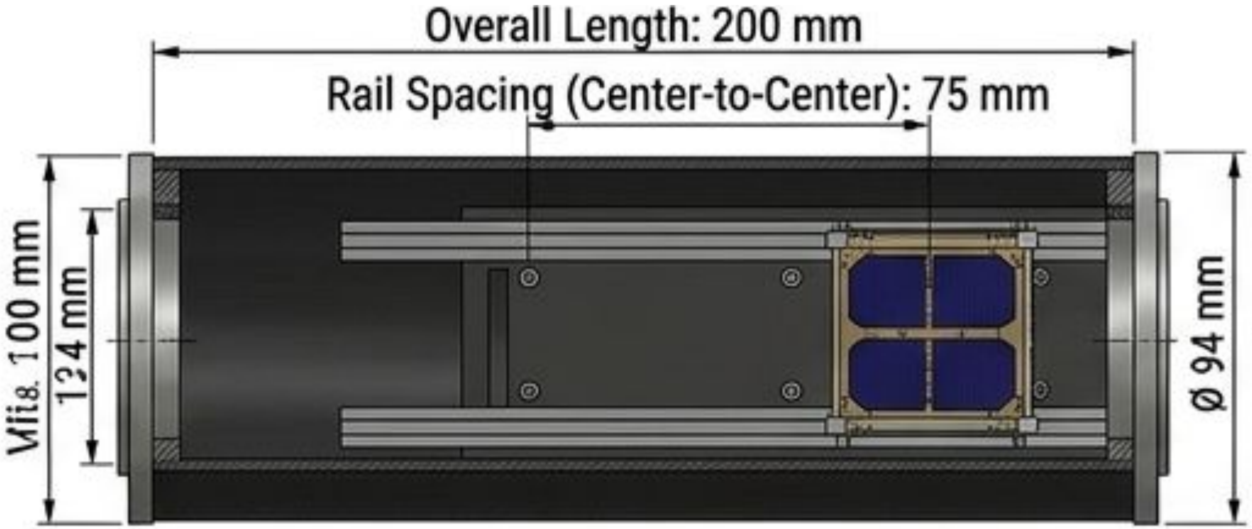


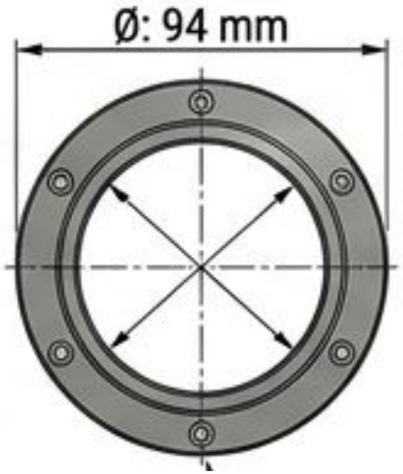
Payload Bay & CAN-7USAT Ejection Overview

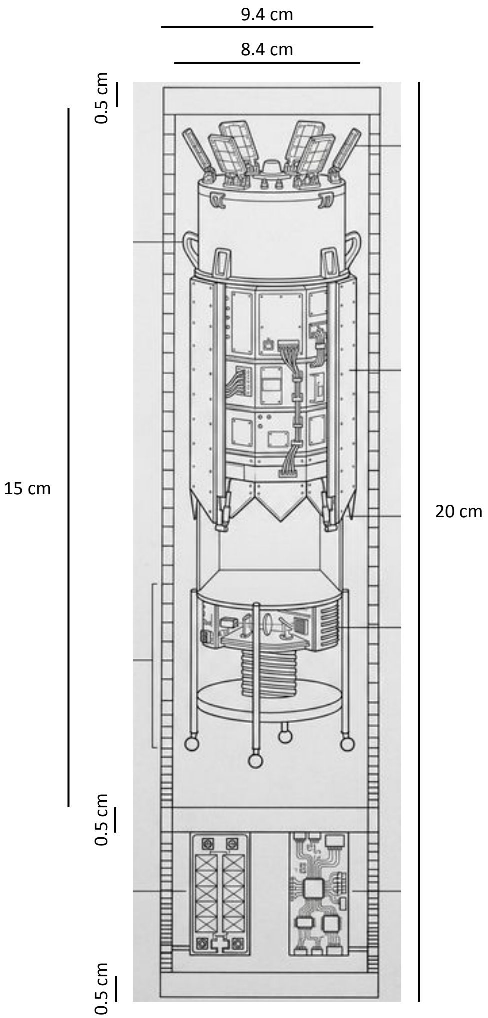


Schematic Outline









CANSAT-7USAT Frame with Drag Decelerator

CAN-7USAT Frame

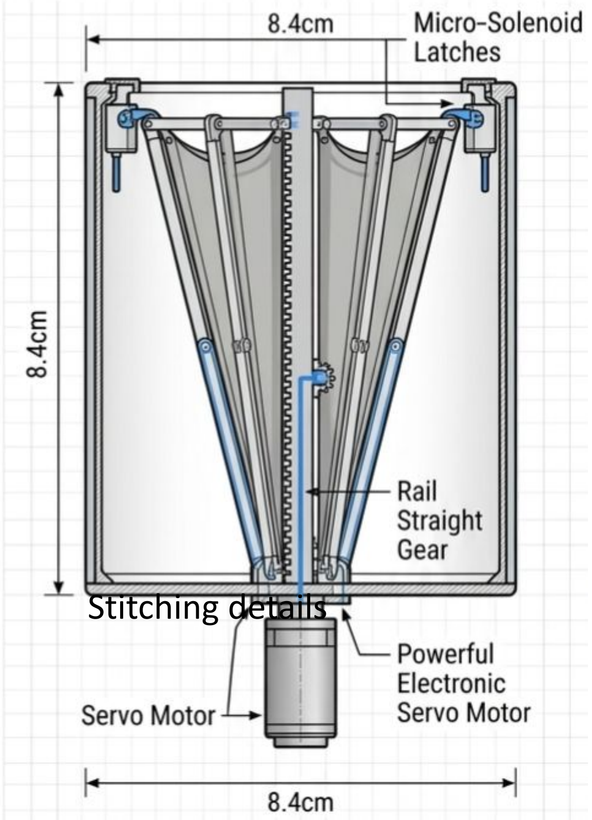
System Overview

The Deployable Aerodynamic Decelerator (DAD) utilizes an inverted conical flare configurative decelerator that is structurally integrated at the aft-end of a ballistic capsule assembly. It is engineered as a rigid deployable decelerator to bypass the volumetric limits of standard launch shrouds. The system deploys by transitioning from a stowed, low-drag cylinder into an expanded geometric surface, which rapidly maximizes the reference drag area (A).

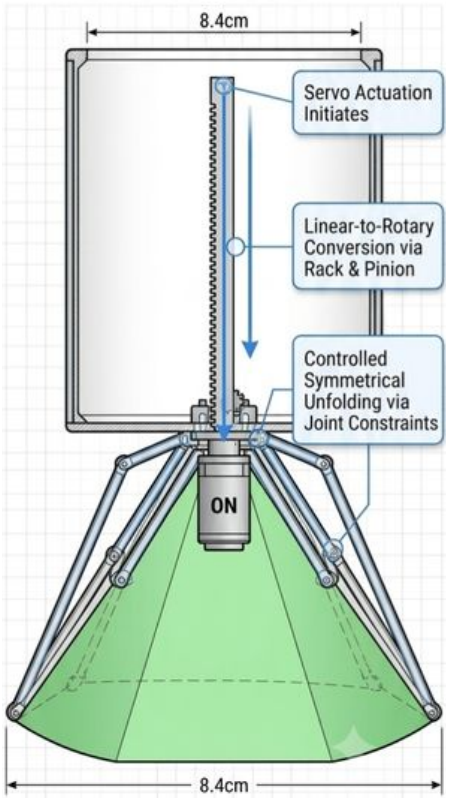
- **Deployment Mechanism:** To actuate this transition, a rail straight gear component is placed at the center at the bottom of the conical funnel. Once the Servo motor is turned ON, the funnel structure frame is folded downward.

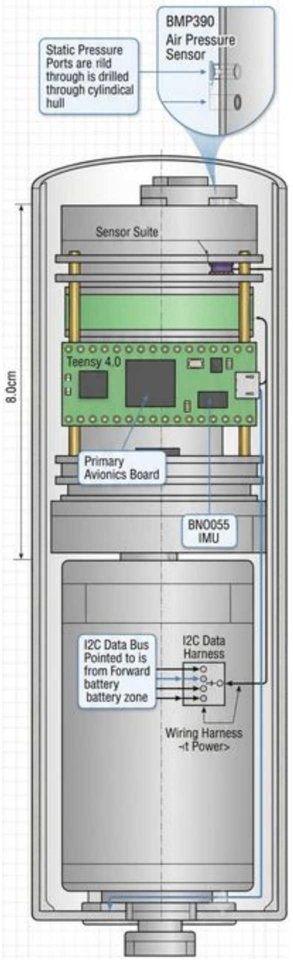
CANSAT-7USAT & Drag Decelerator Overview

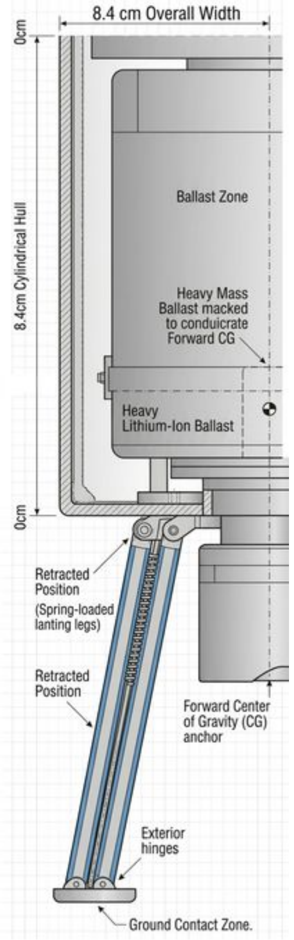
STOWED CONFIGURATION

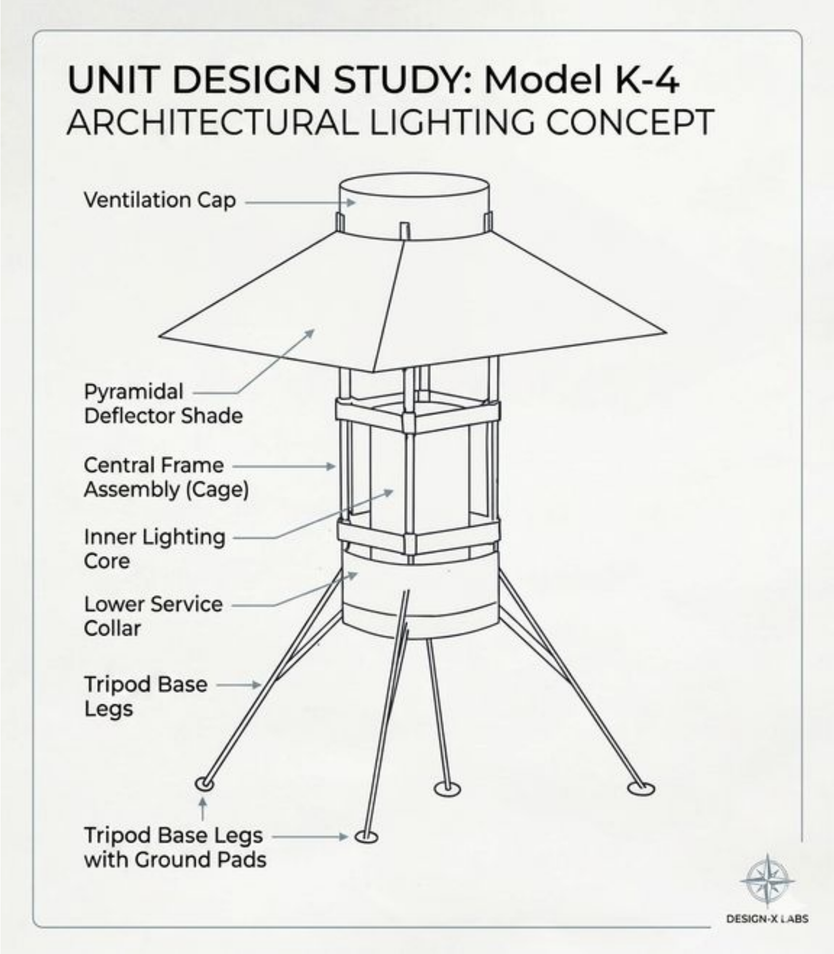


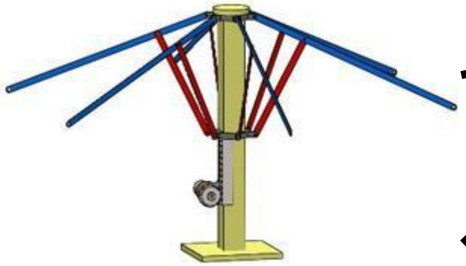
DEPLOYED CONFIGURATION

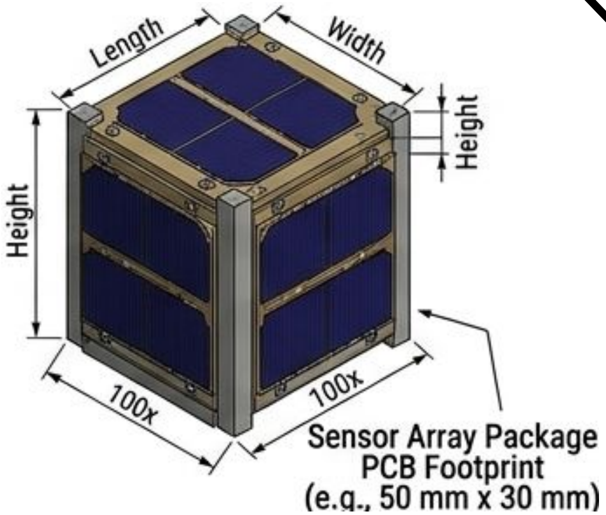


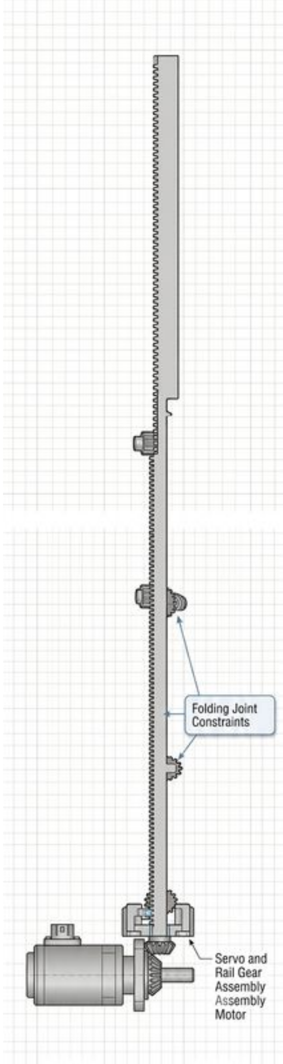


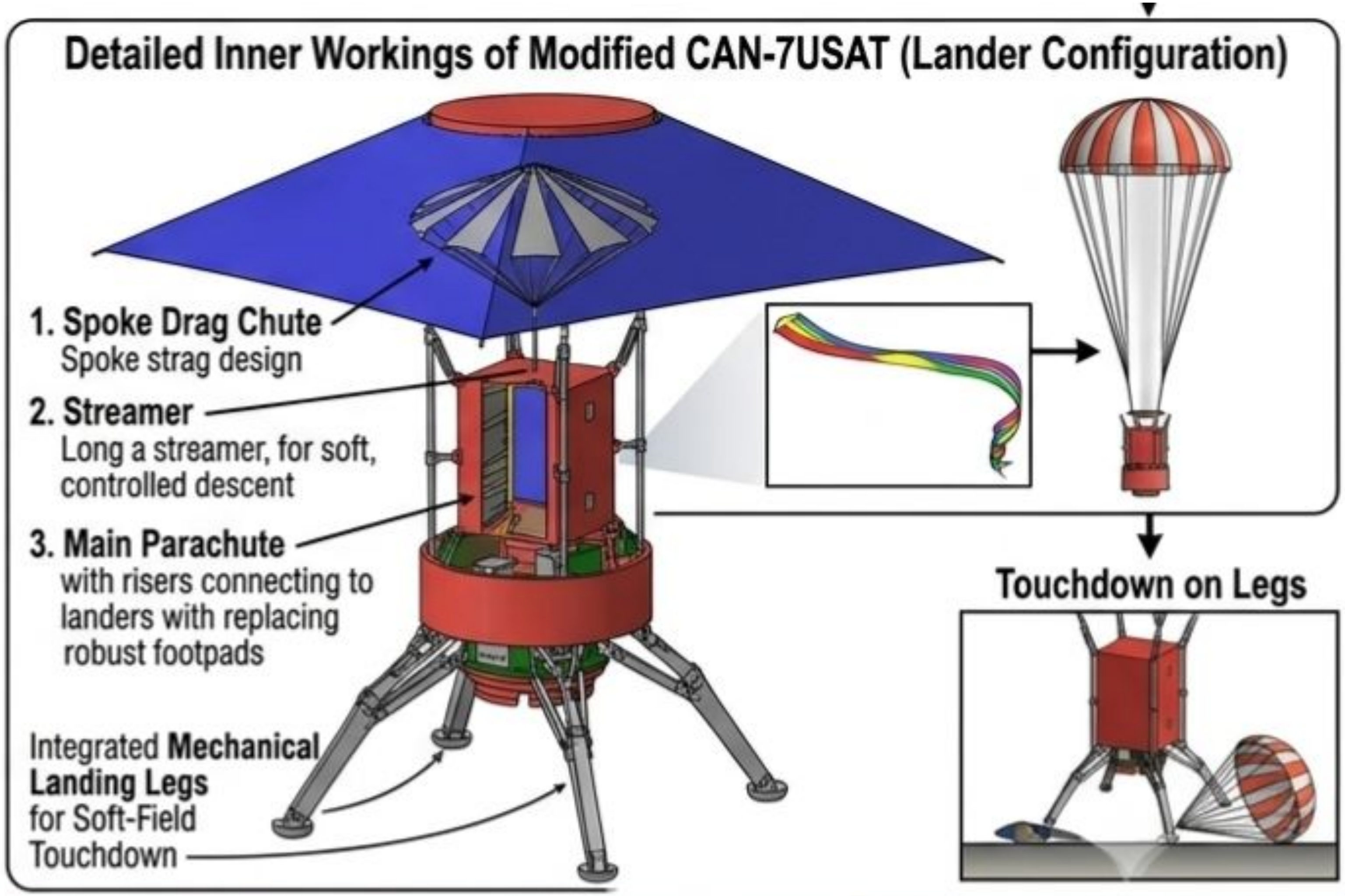



UNIT DESIGN STUDY: Model K-4
ARCHITECTURAL LIGHTING CONCEPT
Ventilation Cap
Pyramidal Deflector Shade
Central Frame Assembly (Cage)
Inner Lighting Core
Lower Service Collar
Tripod Base Legs
Tripod Base Legs with Ground Pads
DESIGN-X LABS




Length
Width
Height
Height
100x
100x
Sensor Array Package PCB Footprint (e.g., 50 mm x 30 mm)


Folding Joint Constraints
Servo and Rail Gear Assembly Assembly Motor



Detailed Inner Workings of Modified CAN-7USAT (Lander Configuration)
1. Spoke Drag Chute
Spoke strag design
2. Streamer
Long a streamer, for soft, controlled descent
3. Main Parachute
with risers connecting to landers with replacing robust footpads
Integrated Mechanical Landing Legs for Soft-Field Touchdown
Touchdown on Legs